Chapter 11 Relativistic Kinematics

1. Galilean Relativity

A Reference frame is a set of the origin and axes that forms a coordinate of space and time for describing the dynamics of a system. Inertial frames are the reference frames that move with constant velocities with respect to each other. Two basic postulates of Galilean relativity are

- The Newton's laws are invariant in inertial frames.
- The space geometry of physics is the Euclidian space and the Euclidian distance

$$dr^2 = dx^2 + dy^2 + dx^2 \tag{1}$$

where $\vec{r} = (x, y, z)$, is invariant under physics transformations. Consider two inertial frames, K and K', K' is moving with a constant velocity $\vec{v}$ with respect to K. The Galilean transformation between K and K' is simply

$$\vec{r}\,' = \vec{r} - \vec{v}\,t \qquad \text{and} \qquad t' = t \tag{2}$$

1.1. An Example of Invariance of the Newton's Laws with Galilean Relativity

For a particle moving in one dimension under an interaction field of potential V, the Newton's 2nd law in the reference frame K' of (x', t') is

$$\frac{dp'_x}{dt'} = -\frac{\partial V(x', t')}{\partial x'} \qquad \text{where} \qquad p'_x = m\frac{dx'}{dt'} \tag{3}$$

Now we transfer Eq. (3) into the reference frame K using the Galilean transformation in Eq. (2). With $x = x' + vt'$, $t = t'$,

$$\frac{d}{dt'} = \frac{d}{dt} \qquad \text{and} \qquad \frac{\partial}{\partial x'} = \frac{\partial x}{\partial x'} \cdot \frac{\partial}{\partial x} + \frac{\partial t}{\partial x'} \cdot \frac{\partial}{\partial t} = \frac{\partial}{\partial x}\,, \tag{4}$$

the derivates with prime variables can be converted as

$$p'_x = m\frac{dx'}{dt'} = m\frac{d(x - vt)}{dt} = p_x - mv \tag{5}$$

where $p_x = m(dx/dt)$ and

$$\frac{dp'_x}{dt'} = \frac{dp_x}{dt} \qquad \text{and} \qquad \frac{\partial V(x')}{\partial x'} = \frac{\partial V(x - vt)}{\partial x}$$

The equation of motion in the reference frame K is therefore in the same form of that in the reference frame K',

$$\frac{dp_x}{dt} = -\frac{\partial V(x, t)}{\partial x}$$

1.2. Two Contradictions to Galilean Relativity

- Experimentally, the speed of light was found to be a constant in different reference frames for example, along different direction on the Earth.
- Maxwell equations are not invariant with the Galilean transformation.

The Transformation of the Wave Equation with Galilean Relativity

Consider the wave equation for one-dimensional electromagnetic wave in the reference frame K',

$$\left(\frac{\partial^2}{\partial x'^2} - \frac{1}{c^2}\frac{\partial^2}{\partial t'^2}\right)\Psi(x', t') = 0 \tag{6}$$

With the Galilean transformation from the reference frame K' to K,

$$x = x' + vt' \qquad \text{and} \qquad t = t'$$

the transformation of the wave equation can be obtained as

$$\frac{\partial}{\partial x'} = \frac{\partial x}{\partial x'}\frac{\partial}{\partial x} + \frac{\partial t}{\partial x'}\frac{\partial}{\partial t} = \frac{\partial}{\partial x} \qquad \Longrightarrow \qquad \frac{\partial^2}{\partial x'^2} = \frac{\partial^2}{\partial x^2}$$

$$\frac{\partial}{\partial t'} = \frac{\partial x}{\partial t'}\frac{\partial}{\partial x} + \frac{\partial t}{\partial t'}\frac{\partial}{\partial t} = v\frac{\partial}{\partial x} + \frac{\partial}{\partial t} \qquad \Longrightarrow \qquad \begin{aligned}\frac{\partial^2}{\partial t'^2} &= \left(v\frac{\partial}{\partial x} + \frac{\partial}{\partial t}\right)\left(v\frac{\partial}{\partial x} + \frac{\partial}{\partial t}\right) \\ &= \frac{\partial^2}{\partial t^2} + 2v\frac{\partial^2}{\partial x \partial t} + v^2\frac{\partial^2}{\partial x^2}\end{aligned}$$

The wave equation in reference frame K is therefore

$$\left[\left(1 - \frac{v^2}{c^2}\right)\frac{\partial^2}{\partial x^2} - \frac{1}{c^2}\left(\frac{\partial^2}{\partial t^2} + 2v\frac{\partial^2}{\partial x \partial t}\right)\right]\Psi \;=\; 0 \tag{7}$$

which is different from the wave equation for an electromagnetic wave in reference frame K'.

1.3. Invariance of the Maxwell Equations with Lorentz Transformation

Before the discovery of the Einstein relativity, Lorentz and Poincaré discovered that the Maxwell equations are invariant under the Lorentz transformation. The Lorentz transformation from the reference frame K' to K for one dimensional space is

$$ct = \gamma\,(ct' + \beta x') \qquad \text{and} \qquad x = \gamma\,(x' + \beta ct') \tag{8}$$

The transformation of the wave equation with the Lorentz transformation is

$$\frac{\partial}{\partial x'} = \frac{\partial x}{\partial x'}\frac{\partial}{\partial x} + \frac{\partial t}{\partial x'}\frac{\partial}{\partial t} = \gamma\left(\frac{\partial}{\partial x} + \frac{\beta}{c}\frac{\partial}{\partial t}\right) \quad \Longrightarrow \quad \begin{aligned}\frac{\partial^2}{\partial x'^2} &= \gamma^2\left(\frac{\partial}{\partial x} + \frac{\beta}{c}\frac{\partial}{\partial t}\right)\left(\frac{\partial}{\partial x} + \frac{\beta}{c}\frac{\partial}{\partial t}\right) \\ &= \gamma^2\left(\frac{\partial^2}{\partial x^2} + \frac{2\beta}{c}\frac{\partial^2}{\partial x \partial t} + \frac{\beta^2}{c^2}\frac{\partial^2}{\partial t^2}\right)\end{aligned}$$

$$\frac{\partial}{\partial t'} = \frac{\partial x}{\partial t'}\frac{\partial}{\partial x} + \frac{\partial t}{\partial t'}\frac{\partial}{\partial t} = \gamma\left(\beta c\frac{\partial}{\partial x} + \frac{\partial}{\partial t}\right) \quad \Longrightarrow \quad \begin{aligned}\frac{\partial^2}{\partial t'^2} &= \gamma^2\left(v\frac{\partial}{\partial x} + \frac{\partial}{\partial t}\right)\left(v\frac{\partial}{\partial x} + \frac{\partial}{\partial t}\right) \\ &= \gamma^2\left(\frac{\partial^2}{\partial t^2} + 2v\frac{\partial^2}{\partial x\partial t} + v^2\frac{\partial^2}{\partial x^2}\right)\end{aligned}$$

and

$$\begin{aligned}\left(\frac{\partial^2}{\partial x'^2} - \frac{1}{c^2}\frac{\partial^2}{\partial t'^2}\right)\Psi &= \left[\gamma^2\left(\frac{\partial^2}{\partial x^2} + \frac{2\beta}{c}\frac{\partial^2}{\partial x\partial t} + \frac{\beta^2}{c^2}\frac{\partial^2}{\partial t^2}\right) - \frac{\gamma^2}{c^2}\left(\frac{\partial^2}{\partial t^2} + 2v\frac{\partial^2}{\partial x\partial t} + v^2\frac{\partial^2}{\partial x^2}\right)\right]\Psi \\ &= \left(\frac{\partial^2}{\partial x^2} - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\right)\Psi\end{aligned}$$

The wave equation (and the Maxwell equations) is therefore invariant under the Lorentz transformation. The Lorentz transformation could thus be the candidate to replace the Galilean transformation for a new unified relativity of the mechanics and electrodynamics. With the invariance of the Maxwell equations, the speed of electromagnetic waves in vacuum has to be a common constant in all the inertial frames if the Maxwell equations are correct. The experimental conformation of the Maxwell equations and the observation of a constant speed of light in vacuum convinced Einstein to develop the Einstein relativity.

2. Einstein Special Relativity

(a) Two Basic Postulates of the Special Relativity

I. The laws of nature are identical in all inertial frames.

II. The speed of light in vacuum is a constant to all observers in all inertial frames.

(b) Space Geometry of Special Relativity

The space consistent with the Einstein relativity is four-dimensional Minkowski space $(x_0, \vec{r}) = (ct, x, y, z)$, where c is the speed of light in vacuum and $x_0 = ct$ is another coordinate in additional to $\vec{r}$ in the Minkowski space. Each point in the Minkowski space is an event occurred at location (x, y, z) and at time t. The distance in the Minkowski space is defined as

$$ds^2 = dx_0^2 - dx^2 - dy^2 - dz^2 \tag{9}$$

which is the distance between two events separated in space and/or time and is the scalar (inner) product of the space vector in Minkowski space,

$$ds^2 = \begin{pmatrix} dx_0 & dx & dy & dz \end{pmatrix} \mathbf{g} \begin{pmatrix} dx_0 \\ dx \\ dy \\ dz \end{pmatrix} \tag{10}$$

where $\mathbf{g}$ is the metric tensor of the Minkowski space

$$\mathbf{g} = \begin{pmatrix} 1 & & & 0 \\ & -1 & & \\ & & -1 & \\ 0 & & & -1 \end{pmatrix} \qquad \text{with} \qquad \mathbf{g}^{-1} = \mathbf{g} \tag{11}$$

3. Lorentz Transformation

In general, the four-dimensional Minkowski space can be considered as homogeneous and isotropic. The transformation among inertial frames of the Minkowski space has to be linear (Ref: HW 11.1) and scalar invariant.

Scalar Invariance of Transformations in Minkowski Space

Consider a transformation between two reference frame K and K'. The distance between two events in the Minkowski space is

$$ds^2 = dx_0^2 - dx^2 - dy^2 - dz^2$$

in K and

$$ds'^2 = dx_0'^2 - dx'^2 - dy'^2 - dz'^2$$

in K', respectively. The scalar invariance of the transformation in the Minkowski space requires

$$ds^2 = ds'^2 \tag{12}$$

3.1. Lorentz Transformation for $\vec{v} \parallel \vec{e}_x$

Consider that K' is moving in the x direction with a velocity v with respect to K and the origins of two frames are coincident at $t = 0$. In the directions perpendicular to $\vec{v} = v\vec{e}_x$,

$$y' = y \qquad \text{and} \qquad z' = z \tag{13}$$

In the direction along $\vec{v} = v\vec{e}_x$, the possible linear transformation between K and K' is

$$\left\{ \begin{array}{lcl} x_0' & = & mx_0 + nx \\ x' & = & px_0 + qx \end{array} \right. \tag{14}$$

where m, n, p, and q are four constant coefficients. For $ds^2 = ds'^2$, we have

$$\begin{aligned} x_0^2 - x^2 &= x_0'^2 - x'^2 \\ &= (mx_0 + nx)^2 - (px_0 + qx)^2 \\ &= (m^2 - p^2)x_0^2 + (n^2 - q^2)x^2 + 2(mn - pq)x_0 x \end{aligned}$$

For arbitrary x_0 and x, we have

$$\begin{cases} m^2 - p^2 = \ 1 \\ n^2 - q^2 = -1 \\ nm - pq = 0 \end{cases} \qquad \Longrightarrow \qquad (1+p^2)(q^2-1) = m^2 n^2 = p^2 q^2$$

which yields

$$q^2 - p^2 = 1 \qquad \text{and} \qquad m = q\,, \quad n = p \tag{15}$$

To determine p and q, we consider the location of the origin of K' in the both reference frames. The origin of K' is at $x' = 0$ in K' and at $x = vt$ in K, respectively. The transformation of Eq. (14) for the location of the origin of K' is therefore

$$0 = px_0 + qx = pct + qvt \qquad \Longrightarrow \qquad \frac{p}{q} = -\frac{v}{c} = -\beta$$

which yields

$$1 - \beta^2 = 1 - \frac{p^2}{q^2} = \frac{q^2 - p^2}{q^2} = \frac{1}{q^2} \qquad \text{and} \qquad q = \frac{1}{\sqrt{1-\beta^2}} = \gamma$$

With $q = \gamma$ and $p = -\beta\gamma$, the Lorentz transformation for $\vec{v} \parallel \vec{e}_x$ can be written as

$$\begin{cases} x'_0 &= \gamma(x_0 - \beta x) \\ x' &= \gamma(x - \beta x_0) \end{cases} \qquad \text{and} \qquad \begin{cases} x_0 &= \gamma(x'_0 + \beta x') \\ x &= \gamma(x' + \beta x'_0) \end{cases} \tag{16}$$

where the inverse transformation is obtained by switching the direction of v, *i.e.* $\beta \to -\beta$. In the case of $c >> v$, $\beta << 1$, $\gamma \simeq 1$ and the Lorentz transformation becomes

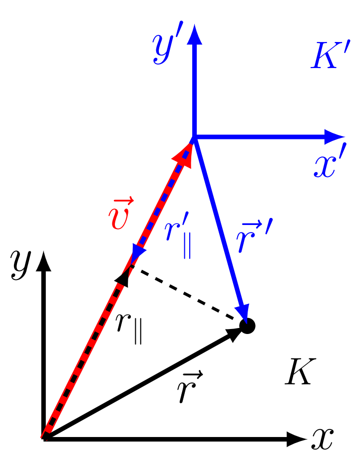
$$t' = t \qquad \text{and} \qquad x' = x - vt$$

which is the Galilean transformation.

3.2. Lorentz Transformation for $\vec{v}$ along Arbitrary Direction

In general, the velocity $\vec{v}$ between K and K' is along an arbitrary direction, but we still can choose coordinates in such a way that the coordinate axes of K and K' are parallel with each other, *i.e.* $\vec{e}_x \parallel \vec{e}_{x'}$, $\vec{e}_y \parallel \vec{e}_{y'}$, and $\vec{e}_z \parallel \vec{e}_{z'}$. Let

$$\begin{aligned} r_\parallel &= \text{the component of } \vec{r} \text{ parallel to } \vec{v} \\ \vec{r}_\perp &= \text{the components of } \vec{r} \text{ perpendicular to } \vec{v} \end{aligned}$$



With $\vec{r}$ being projected into the parallel and perpendicular directions, the 1D Lorentz transformation in Eq. (16) can be directly applied as

$$\begin{cases} x'_0 &= \gamma(x_0 - \beta r_\parallel) \\ r'_\parallel &= \gamma(r_\parallel - \beta x_0) \\ \vec{r}'_\perp &= \vec{r}_\perp \end{cases} \tag{17}$$

Since the parallel and perpendicular components of $\vec{r}$ can be calculated as

$$r_\parallel = \frac{\vec{r}\cdot\vec{\beta}}{\beta} \qquad \text{and} \qquad \vec{r}_\perp = \vec{r} - \frac{\vec{\beta}}{\beta}\left(\frac{\vec{r}\cdot\vec{\beta}}{\beta}\right)$$

the Lorentz transformation in Eq. (17) can be rewritten as

$$\begin{cases} x'_0 &= \gamma\left(x_0 - \beta\,\frac{\vec{r}\cdot\vec{\beta}}{\beta}\right) \\ \frac{\vec{r}'\cdot\vec{\beta}}{\beta} &= \gamma\left(\frac{\vec{r}\cdot\vec{\beta}}{\beta} - \beta x_0\right) \end{cases} \Longrightarrow \begin{cases} x'_0 &= \gamma\left(x_0 - \vec{\beta}\cdot\vec{r}\right) \\ \frac{\vec{\beta}}{\beta^2}\left(\vec{\beta}\cdot\vec{r}'\right) &= \gamma\,\frac{\vec{\beta}}{\beta^2}\left(\vec{r}\cdot\vec{\beta} - \beta^2 x_0\right) \end{cases} \tag{18}$$

and in the directions that is perpendicular to $\vec{v}$

$$\vec{r}' - \frac{\vec{\beta}}{\beta^2}\left(\vec{\beta}\cdot\vec{r}'\right) = \vec{r} - \frac{\vec{\beta}}{\beta^2}\left(\vec{\beta}\cdot\vec{r}\right) \tag{19}$$

Combining both the directions together yields the general Lorentz transformation,

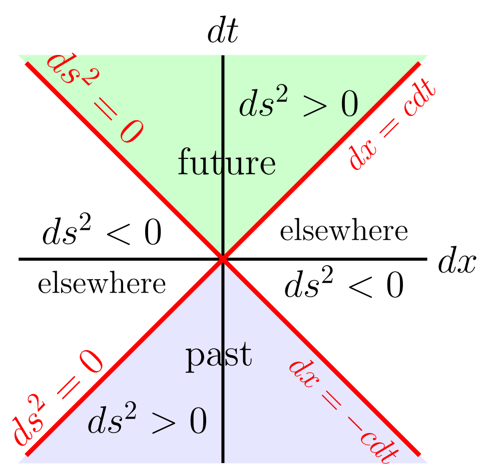
$$\begin{cases} x'_0 &= \gamma\left(x_0 - \vec{\beta}\cdot\vec{r}\right) \\ \vec{r}' &= \vec{r} + \frac{(\gamma-1)}{\beta^2}\left(\vec{\beta}\cdot\vec{r}\right)\vec{\beta} - \gamma\vec{\beta}x_0 \end{cases} \tag{20}$$

3.3. Light Cone

Since the distance between events in the Minkowski space,

$$ds^2 = dx_0^2 - (dx^2 + dy^2 + dz^2)$$

is invariant, the Minkowski space is divided into four regions that cannot be connected by Lorentz transformations.



(a) $ds^2 = 0$ — Lightlike Interval

$ds^2 = 0$ are 45° diagonal lines (surface in 4D space) between the time and space axes in the Minkowski space. Any two events on these surfaces are connected by light in all inertial frames,

$$dx_0'^2 - (dx'^2 + dy'^2 + dz'^2) = dx_0^2 - (dx^2 + dy^2 + dz^2) = 0$$

These lightlike surfaces divide the Minkowski space into four cone shaped regions with either $ds^2 > 0$ or $ds^2 < 0$. Events in any of these four regions cannot be transferred into another region by any Lorentz transformation because of the invariance of ds^2 and the continuity of the Lorentz transformation for possible crossing the lines of $ds^2 = 0$.

(b) $ds^2 > 0$ — Timelike Interval

With $ds^2 > 0$, it is always possible to find a K' frame such that

$$ds'^2 = dx_0'^2 = ds^2 > 0 \qquad \text{and} \qquad dx'^2 + dy'^2 + dz'^2 = 0$$

The two events in K' can therefore occur at the same location, but can never occur at the same time. If $dx_0 > 0$, for the continuity of the space, $dx_0' > 0$ and vise versa. The timelike interval is thus divided into two regions with each region containing the events in "past" and events in "future" separately. The region of "past" and the region of "future" are separated by the lightlike interval of $ds^2 = 0$ and cannot be crossed by any Lorentz transformation. In the Minkowski space, the causality holds.

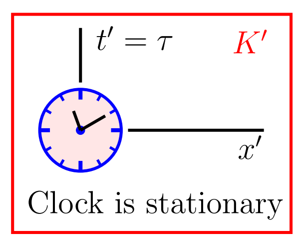
(c) $ds^2 < 0$ — Spacelike Interval

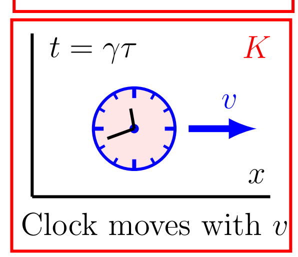
With $ds^2 < 0$, it is always possible to find a K' frame such that

$$ds'^2 = -(dx'^2 + dy'^2 + dz'^2) = ds^2 < 0 \qquad \text{and} \qquad dx_0' = 0$$

Two events in K' can therefore occur at the same time, but different locations and they can never be causally connected, *i.e.* all possibilities of $dt' > 0$, $dt' = 0$ and $dt' < 0$ are permitted. In the spacelike interval, the sequential order of events in the (x, y, z) space cannot be altered, i.e. the spacelike interval is divided into two regions with each region containing events of $dx > 0$ and $dx < 0$, respectively. The two regions are separated by the lightlike interval of $ds^2 = 0$ and cannot be crossed by any Lorentz transformation.

3.4. Proper Time, Time Dilation, and Length Contraction





The proper time τ is defined as the time observed for an object in a reference frame in which the object is stationary.

(a) Time Dilation

Consider a clock is moving in lab frame K with a velocity v along the x direction and frame K' is moving with the clock. In frame K' the clock is stationary, *i.e.* $dx' = 0$ and $t' = \tau$, and in lab frame K the coordinate of the clock is (x, t). The Lorentz transformation between K and K' is

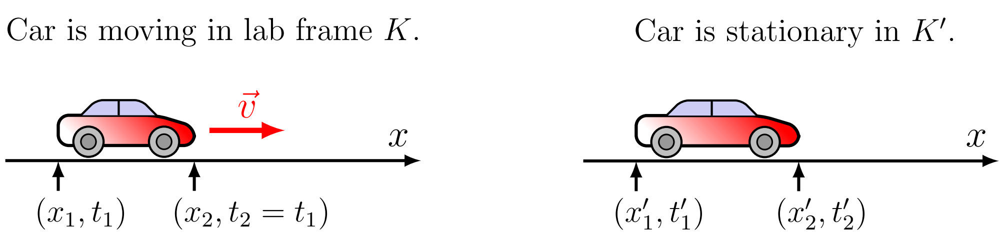
$$\begin{cases} 0 = dx' = \gamma(dx - \beta c dt) \\ c d\tau = \gamma(c dt - \beta dx) \end{cases} \quad \Longrightarrow \quad \begin{array}{l} dx = v dt \\ c d\tau = \gamma(c - v\beta) dt = \dfrac{c}{\gamma} dt \end{array}$$

Therefore, the time observed when riding with a moving clock is

$$dt = \gamma d\tau$$

Since $dt > d\tau$, the time observed in the lab frame dt is longer (faster) than the time observed when moving with the clock $d\tau$, *i.e.* a moving clock goes slower from the lab frame's point of view, which is called the time dilation.

(b) Length Contraction



Consider a car moves with a velocity v along the x direction in lab frame K. Frame K' is moving with the car. In frame K', the length of the car is $dx' = x'_2 - x'_1$, where x'_2 and x'_1 are the coordinates of the front and back end of the car. Since the car is not moving in K', the measurement of x'_1 and x'_2 does not need to be made at a same time, *i.e.* t'_1 and t'_2, the time of the measurement in K', do not need to be the same. To measure the length of the car in the lab frame, the coordinate of the two ends of the car x_1 and x_2 have to be measured simultaneously at a certain time $t_1 = t_2$, where t_1 and t_2 are the time of the measurement of x_1 and x_2, respectively. With $dx = x_2 - x_1$ and $dt = t_2 - t_1 = 0$ in the lab frame,

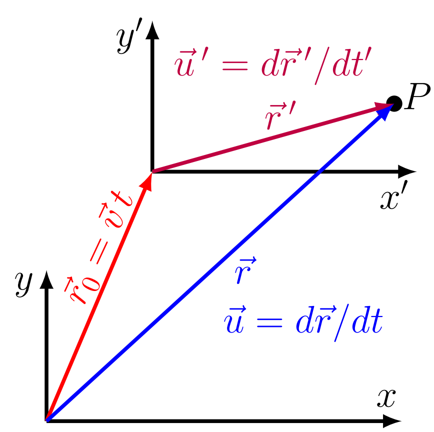
$$dx' = \gamma(dx - \beta c dt) = \gamma dx > dx$$

Therefore, a moving car is shorter when observed from the lab frame.

3.5. Transformation and Addition of Velocities

Consider a reference frame K' is moving with a constant velocity $\vec{v}$ with respect to reference frame K. The velocity of an object observed in K and K' are, respectively,

$$\vec{u} = \frac{d\vec{r}}{dt} \qquad \text{and} \qquad \vec{u}\,' = \frac{d\vec{r}\,'}{dt'}$$



In the case of $\vec{v} \parallel \vec{e}_x$, from the Lorentz transformation between K and K'

$$\left\{ \begin{array}{rcl} cdt & = & \gamma(cdt' + \beta dx') \\ dx & = & \gamma(dx' + \beta cdt') \\ dy & = & dy' \\ dz & = & dz' \end{array} \right.$$

the velocity transformation can be obtained as

$$\begin{cases} u_x &= \dfrac{dx}{dt} = c\dfrac{dx' + \beta c dt'}{cdt' + \beta dx'} = \dfrac{v + u'_x}{1 + vu'_x/c^2} \\ u_y &= \dfrac{dy}{dt} = c\dfrac{dy'}{\gamma(cdt' + \beta dx')} = \dfrac{u'_y}{\gamma(1 + vu'_x/c^2)} \\ u_z &= \dfrac{dz}{dt} = \dfrac{u'_z}{\gamma(1 + vu'_x/c^2)} \end{cases} \tag{21}$$

In general,

$$\begin{cases} u_\parallel &= \dfrac{v + u'_\parallel}{1 + \vec{v} \cdot \vec{u}\,'/c^2} \\ \vec{u}_\perp &= \dfrac{\vec{u}\,'_\perp}{\gamma_v(1 + \vec{v} \cdot \vec{u}\,'/c^2)} \end{cases} \tag{22}$$

where $u_\parallel$ and $\vec{u}_\perp$, as well as $u'_\parallel$ and $\vec{u}'_\perp$, are the components of velocity parallel and perpendicular, respectively, to $\vec{v}$. Equation (22) can also be used for the addition of two velocities, $\vec{v}$ and $\vec{u}$. When $v = |\vec{v}| << c$, Eq. (22) is reduced to the Galilean transformation of $\vec{u} = \vec{v} + \vec{u}\,'$. If $v = c$ and $u' = c$, $u_\parallel = c$ and $\vec{u}_\perp = 0$. Therefore, the maximal speed of an object is c no matter where you observe it.

4. Mathematical Properties of Lorentz Transformations in Minkowski Space

4.1. Three-Dimensional Euclidian Space

In order to better understand the symmetry of transformations in the Minkowski space, we first briefly review the unitary transformations in 3-dimensional Euclidian space. In the Euclidian space,

coordinate:	$\vec{r} = (x_1,\, x_2,\, x_3)$		
vector:	$\vec{u} = (u_1,\, u_2,\, u_3)$		
transformation:	$\vec{r}\,' = \mathbf{A}\,\vec{r}^{\,T}$	and	$\vec{u}\,' = \mathbf{A}\,\vec{u}^{\,T}$

For example, a rotational transformation in the x-y plane is

$$\mathbf{A} = \begin{pmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

In the Euclidian Space, unitary transformations preserve scalar products especially the Euclidian distance, *i.e.* $\vec{r} \cdot \vec{r} = \vec{r}\,' \cdot \vec{r}\,'$ for $\vec{r}\,' = \mathbf{A}\,\vec{r}$ and $\mathbf{A}$ is unitary. Since

$$\begin{aligned} \vec{r}\,' \cdot \vec{r}\,' = \begin{pmatrix} x' & y' & z' \end{pmatrix} \begin{pmatrix} x' & y' & z' \end{pmatrix}^T &= \begin{pmatrix} x & y & z \end{pmatrix} \mathbf{A}^T \mathbf{A} \begin{pmatrix} x & y & z \end{pmatrix}^T \\ &= \begin{pmatrix} x & y & z \end{pmatrix} \begin{pmatrix} x & y & z \end{pmatrix}^T = \vec{r} \cdot \vec{r} \end{aligned}$$

$\mathbf{A}$ is unitary, where

$$\mathbf{A}^T \mathbf{A} = \mathbf{I} \qquad \Longrightarrow \qquad \mathbf{A}^{-1} = \mathbf{A}^T \tag{23}$$

and the determinant of $\mathbf{A}$ is

$$\det(\mathbf{A}^T\mathbf{A}) = (\det\mathbf{A})^2 = 1 \qquad \Longrightarrow \qquad \det\mathbf{A} = \pm 1 \tag{24}$$

For the scalar product of any two vector in the Euclidian space,

$$\vec{V}\,' \cdot \vec{U}\,' \;=\; \vec{V}^T\mathbf{A}^T\mathbf{A}\,\vec{U} \;=\; \vec{V}^T\vec{U} \;=\; \vec{V}\cdot\vec{U}$$

i.e. the scalar products are invariant under the unitary transformation in the Euclidian space.

4.2. Four-Dimensional Minkowski Space

For four-dimensional vectors in the Minkowski space, (V_0, V_1, V_2, V_3) and (U_0, U_1, U_2, U_3), the scalar product is defined as

$$V_0U_0 - (V_1U_1 + V_2U_2 + V_3U_3) \;=\; (\; V_0 \quad V_1 \quad V_2 \quad V_3 \;)\,\mathbf{g}\,(\; U_0 \quad U_1 \quad U_2 \quad U_3 \;)^T \tag{25}$$

where $\mathbf{g}$ is the Minkowski metric. If a transformation $\mathbf{U}' = \mathbf{A}\mathbf{U}$ keeps the scalar invariant,

$$\begin{aligned}
&(\; V_0' \quad V_1' \quad V_2' \quad V_3' \;)\,\mathbf{g}\,(\; U_0' \quad U_1' \quad U_2' \quad U_3' \;)^T \\
&\qquad = (\; V_0 \quad V_1 \quad V_2 \quad V_3 \;)\mathbf{A}^T\mathbf{g}\mathbf{A}(\; U_0 \quad U_1 \quad U_2 \quad U_3 \;)^T \\
&\qquad = (\; V_0 \quad V_1 \quad V_2 \quad V_3 \;)\,\mathbf{g}\,(\; U_0 \quad U_1 \quad U_2 \quad U_3 \;)^T
\end{aligned}$$

Therefore, the transformation that preserves the scalar product in the Minkowski space satisfies

$$\mathbf{A}^T\mathbf{g}\,\mathbf{A} = \mathbf{g} \tag{26}$$

Since

$$\det(\mathbf{A}^T\mathbf{g}\,\mathbf{A}) = (\det\mathbf{g})(\det\mathbf{A})^2 = \det\mathbf{g} = -1$$

the transformations in the Minkowski space also require the determinant of $\mathbf{A}$ to be

$$\det\mathbf{A} = \pm 1 \tag{27}$$

where $\det\mathbf{A} = -1$ is a transformation combined with a mirror transformation and $\det\mathbf{A} = 1$ is a transformation without a mirror transformation. Because of the continuity of the Lorentz transformation, as the Lorentz transformation is reduced to the identity when $\beta \to 0$, the Lorentz transformations requires $\det\mathbf{A} = 1$.

For the general Lorentz transformation obtained in Section 3.2

$$\left\{\begin{aligned}
x_0' &= \gamma\left(x_0 - \vec{\beta}\cdot\vec{r}\right) \\
\vec{r}\,' &= \vec{r} + \frac{(\gamma-1)}{\beta^2}\left(\vec{\beta}\cdot\vec{r}\right)\vec{\beta} - \gamma\vec{\beta}x_0
\end{aligned}\right. \tag{28}$$

the matrix form can be written as

$$\mathbf{A} = \mathbf{A}_1 + \mathbf{A}_2 \tag{29}$$

where

$$\mathbf{A_1} = \begin{pmatrix} \gamma & -\gamma\beta_x & -\gamma\beta_y & -\gamma\beta_z \\ -\gamma\beta_x & 1 & 0 & 0 \\ -\gamma\beta_y & 0 & 1 & 0 \\ -\gamma\beta_z & 0 & 0 & 1 \end{pmatrix} \qquad \text{and} \qquad \mathbf{A_2} = \frac{\gamma-1}{\beta^2}\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & \beta_x^2 & \beta_x\beta_y & \beta_x\beta_z \\ 0 & \beta_x\beta_y & \beta_y^2 & \beta_y\beta_z \\ 0 & \beta_x\beta_z & \beta_y\beta_z & \beta_z^2 \end{pmatrix}$$

Since both $\mathbf{A_1} = \mathbf{A_1}^T$ and $\mathbf{A_2} = \mathbf{A}_2^T$ are symmetric matrix, for the Lorentz transformation,

$$\mathbf{A}^T = \mathbf{A} \qquad \Longrightarrow \qquad \mathbf{AgA} = \mathbf{g} \quad \text{and} \quad \mathbf{A}^{-1} = \mathbf{gAg} \tag{30}$$

4.3. Four-Velocity $(U_0 \,,\, \vec{U}\,)$

Four-dimensional vectors in the Minkowski space are called 4-vector. The velocity vector of an object in the Minkowski space can be defined as

$$\left\{ \begin{aligned} U_0 &= \frac{dx_0}{d\tau} = \frac{dx_0}{dt}\frac{dt}{d\tau} = \gamma_u c \\ \vec{U} &= \frac{d\vec{x}}{d\tau} = \frac{d\vec{x}}{dt}\frac{dt}{d\tau} = \gamma_u \vec{u} \end{aligned} \right. \tag{31}$$

where $\gamma_u = 1/\sqrt{1-u^2/c^2}$ and $\tau = t/\gamma_u$ is the proper time of the object. The transformation of the velocity can be written into the Lorentz transformation of a 4-vector as

$$\mathbf{U'} = \mathbf{AU} \qquad \Longrightarrow \qquad \left\{ \begin{aligned} U_0' &= \gamma_u(U_0 - \beta U_\parallel) \\ \vec{U}' &= \vec{U} - \vec{\beta}\left[\gamma U_0 - \frac{\gamma-1}{\beta^2}(\vec{\beta}\cdot\vec{U})\right] \end{aligned} \right.$$

where $\mathbf{A}$ is the Lorentz transformation matrix in Eq. (29). With $U_\parallel = \vec{U}\cdot\vec{\beta}/\beta$, the transformation of 4-velocity can be rewritten as

$$\left\{ \begin{aligned} U_0' &= \gamma_u(U_0 - \beta U_\parallel) \\ U_\parallel' &= \gamma_u(U_\parallel - \beta U_0) \\ \vec{U}_\perp' &= \vec{U}_\perp \end{aligned} \right. \tag{32}$$

The transformation in Eq. (32) is the same as the one in Eq. (22) in Section 3.5 as the time component of Eq. (32) can be written into

$$\gamma_{u'} = \gamma_u^2\left(1 - \frac{vu_\parallel}{c^2}\right)$$

and the space components of Eq. (32) can then be changed into

$$\left\{ \begin{aligned} u_\parallel' &= \frac{\gamma_u^2(u_\parallel - v)}{\gamma_{u'}} = \frac{u_\parallel - v}{1 - vu_\parallel/c^2} \\ \vec{u}_\perp' &= \frac{\gamma_u\vec{u}_\perp}{\gamma_{u'}} = \frac{\vec{u}_\perp}{\gamma_u(1 - vu_\parallel/c^2)} \end{aligned} \right.$$

It can easily be verified that the scalar product of the 4-velocity is invariant under the Lorentz transformation,

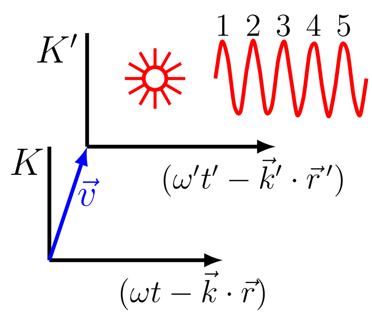
$$U_0^2 - \vec{U}\cdot\vec{U} \;=\; U_0'^2 - \vec{U}'\cdot\vec{U}' \tag{33}$$

4.4. Transformation of Wave Vector $\vec{k}$ and Frequency ω of a Plan Wave and Relativistic Doppler Effect

Consider a plan wave

$$\psi = \psi_0 \sin(\omega t - \vec{k}\cdot\vec{r}) \tag{34}$$

of which the source is stationary in reference frame K' and moves with a constant velocity $\vec{v}$ with respect to reference frame K. When observing the wave from two different reference frames, the time t and space coordinates $\vec{r}$ are different, but the counting of the number of wave oscillation should be the same. Since the number of the oscillation of a wave is determined by the phase of the wave



$$\omega t - \vec{k}\cdot\vec{r}$$

The phase of the wave is invariant under the Lorentz transformation, *i.e.*

$$\omega t - \vec{k}\cdot\vec{r} \;=\; \omega' t' - \vec{k}\,'\cdot\vec{r}\,' \tag{35}$$

where $(\omega, \vec{k}\,)$ and $(\omega', \vec{k}\,')$ are the frequency and wave vector in reference frame K and K', respectively. Let $k_0 = \omega/c$ and $(k_0, \vec{k})$ be the 4-vector of wave vector. The invariance of the phase of a wave in Eq. (35) can be written as a scalar product between the 4-wave-vector and 4-coordinate,

$$(k_0\,,\,\vec{k}\,)\,\mathbf{g}\,(x_0\,,\,\vec{r}\,)^T = (k_0'\,,\,\vec{k}\,')\,\mathbf{g}\,(x_0'\,,\,\vec{r}\,')^T \tag{36}$$

Since $\mathbf{A}^T\mathbf{g}\mathbf{A} = \mathbf{g}$ and the Lorentz transformation of $(x_0\,,\,\vec{r}\,)^T$ is

$$(x_0'\,,\,\vec{r}\,')^T = \mathbf{A}(x_0\,,\,\vec{r}\,)^T \tag{37}$$

where $\mathbf{A}$ is the Lorentz transformation, the left-hand side of Eq. (36) can be written as

$$(k_0\,,\,\vec{k}\,)\,\mathbf{g}\,(x_0\,,\,\vec{r}\,)^T = (k_0\,,\,\vec{k}\,)\,\mathbf{A}^T\mathbf{g}\mathbf{A}\,(x_0\,,\,\vec{r}\,)^T = (k_0\,,\,\vec{k}\,)\,\mathbf{A}^T\mathbf{g}\,(x_0'\,,\,\vec{r}\,')^T \tag{38}$$

The equality between the right-hand sides of Eq. (36) and Eq. (38) requires

$$(k_0'\,,\,\vec{k}\,') = (k_0\,,\,\vec{k}\,)\mathbf{A}^T \qquad \text{or} \qquad (k_0'\,,\,\vec{k}\,')^T = \mathbf{A}(k_0\,,\,\vec{k}\,)^T \tag{39}$$

Therefore, $(k_0\,,\,\vec{k}\,)$ is the correct form of 4-vector for the wave vector in the Minkowski space and, by comparing Eqs. (37) and (39), the Lorentz transformation of $(k_0, \vec{k}\,)$ should be the same as the Lorentz transformation of $(x_0, \vec{r})$. The transformation of the frequency and wavelength can thus be obtained by modifying Eq. (17) as

$$\left\{ \begin{array}{lcl} k'_0 & = & \gamma(k_0 - \vec{\beta}\cdot\vec{k}\,) \\ k'_\parallel & = & \gamma(k_\parallel - \beta k_0) \\ \vec{k}\,'_\perp & = & \vec{k}_\perp \qquad \Longrightarrow \qquad |\vec{k}\,'_\perp| \;=\; |\vec{k}_\perp| \end{array} \right. \tag{40}$$

For an electromagnetic wave,

$$\omega = c|\vec{k}| \qquad \text{and} \qquad k_0 = \omega/c = |\vec{k}| \tag{41}$$

Let θ and θ' be the angle of $\vec{k}$ and $\vec{k}\,'$ relative to $\vec{\beta}$, respectively, *i.e.*

$$\left\{ \begin{array}{lcl} \vec{\beta}\cdot\vec{k} & = & \beta k_0 \cos\theta \\ |\vec{k}_\perp| & = & \beta k_0 \sin\theta \end{array} \right. \qquad \text{and} \qquad \left\{ \begin{array}{lcl} \vec{\beta}\cdot\vec{k}' & = & \beta k'_0 \cos\theta' \\ |\vec{k}'_\perp| & = & \beta k'_0 \sin\theta' \end{array} \right.$$

For electromagnetic waves, the Lorentz transformation of 4-wave-vector $(k_0, \vec{k}\,)$ in Eq. (40) can be written as

$$\left\{ \begin{array}{lcl} \omega' & = & \gamma\omega(1-\beta\cos\theta) \\ \omega'\cos\theta' & = & \gamma\omega(\cos\theta - \beta) \\ \omega'\sin\theta' & = & \omega\sin\theta \end{array} \right. \quad \Longrightarrow \quad \left\{ \begin{array}{lcl} \omega & = & \dfrac{\omega'}{\gamma(1-\beta\cos\theta)} \\ \tan\theta' & = & \dfrac{\sin\theta}{\gamma(\cos\theta-\beta)} \end{array} \right. \tag{42}$$

Let $\omega_0 = \omega'$ is the nature frequency of the electromagnetic wave observed in a reference frame in which the source is not moving. The Doppler shift of the observed frequency of an electromagnetic wave from a moving source is then

$$\omega = \frac{\omega_0}{\gamma(1-\beta\cos\theta)} \tag{43}$$

where for a source moving toward an observer, $\theta = 0$,

$$\omega \;=\; \frac{\omega_0}{\gamma(1-\beta)} \;=\; \gamma(1+\beta)\omega_0 \;\overset{\gamma>>1}{=}\; 2\gamma\omega_0 \qquad \text{for } \vec{k} \parallel \vec{\beta} \tag{44}$$

and for a source moving away from an observer, $\theta = \pi$,

$$\omega \;=\; \frac{\omega_0}{\gamma(1+\beta)} \;\overset{\gamma>>1}{=}\; \frac{\omega_0}{2\gamma} \qquad \text{for } \vec{k} \parallel -\vec{\beta} \tag{45}$$

5. Four-Vector for Momentum and Kinetic Energy

The momentum and kinetic energy defined in the Newtonian mechanics are consistent with the Galilean relativity but incompatible with the Lorentz transformation. A relativistic generalization of the momentum and energy is therefore needed. For $v << c$, the relativistic momenta and energy should go back to the original Newtonian definition of

$$\vec{p} = m\vec{u} \qquad \text{and} \qquad E = E(0) + \frac{1}{2}mu^2$$

In general, we assume the momentum and energy can be written as

$$\vec{p} = \mathcal{M}(u)\vec{u} \qquad \text{and} \qquad E = \mathcal{E}(u) \tag{46}$$

where $\mathcal{M}(u)$ and $\mathcal{E}(u)$ are two functions of the velocity $u = |\vec{u}|$, and

$$\mathcal{M}(0) = m \qquad \text{and} \qquad \left.\frac{d^2\mathcal{E}}{du^2}\right|_{u=0} = \frac{m}{2} \tag{47}$$

To determine the proper form of $\mathcal{M}$ and $\mathcal{E}$, we consider an elastic collision of two identical particles, particle 1 and 2.

5(a) In Center-of-Mass Frame K' of Two Particles

In the center-of-mass (CM) frame K', before the collision, the velocities of two particles have the same magnitude and opposite direction. Let $(\vec{u}_1\,,\,\vec{u}_2) = (\vec{u}\,,\,-\vec{u})$ and $(\vec{u}\,'_1\,,\,\vec{u}\,'_2)$ be the velocities of the particles in K' frame before and after the collision, respectively, *i.e.*

$$\begin{array}{ccc} \text{before collision} & & \text{after collision} \\ (\vec{u}_1\,,\,\vec{u}_2) = (\vec{u}\,,\,-\vec{u}) & \overset{\text{collision}}{\Longrightarrow} & (\vec{u}\,'_1\,,\,\vec{u}\,'_2) \end{array} \tag{48}$$

The conservation of the momentum and energy during the elastic collision in K' frame requires

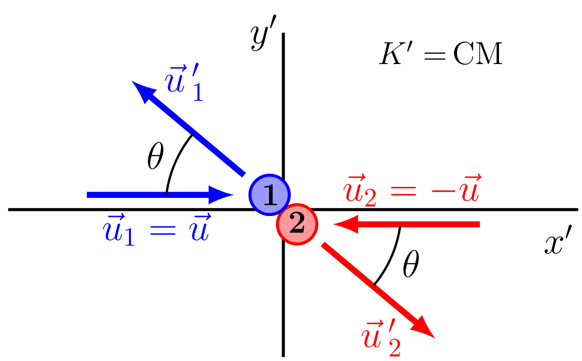
$$\begin{aligned} 0 \;=\; \mathcal{M}(u)\vec{u} - \mathcal{M}(u)\vec{u} \;&=\; \mathcal{M}(u'_1)\vec{u}\,'_1 + \mathcal{M}(u'_2)\vec{u}\,'_2 \\ \mathcal{E}(u) + \mathcal{E}(u) \;&=\; \mathcal{E}(u'_1) + \mathcal{E}(u'_2) \end{aligned}$$

Because two particles are identical and have the same energy before the collision in the CM frames, they should also have the same energy after the collision in the CM frames. If they have different energy after the collision there is no logistic reason to pick which particle has which energy. Therefore,

$$\mathcal{E}(u'_1) = \mathcal{E}(u'_2) = \mathcal{E}(u)$$

which yields

$$\begin{cases} u'_1 \;=\; u'_2 \;=\; u \\ \mathcal{M}(u'_1) = \mathcal{M}(u'_2) = \mathcal{M}(u) \end{cases}$$



The conservation of the momentum is then

$$\mathcal{M}(u)\vec{u}\,'_1 + \mathcal{M}(u)\vec{u}\,'_2 = 0$$

which yields

$$\vec{u}\,'_1 = -\vec{u}\,'_2$$

Since the coordinate axes of reference frame K' can be oriented in any direction, we can choose the collision occurs in the x'-y' plane with $\vec{u}_1 = -\vec{u}_2 = \vec{u}$ parallel with the x' axis.

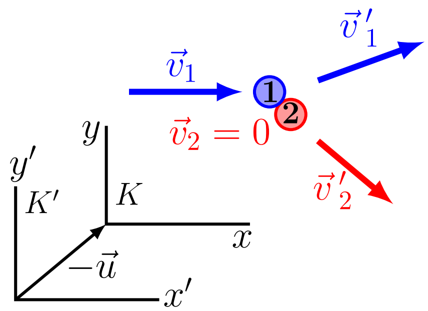
Let θ be the angle between $\vec{u}'_1$ and $\vec{e}_{x'}$ as well as the angle between $\vec{u}'_2$ and $-\vec{e}_{x'}$. The velocity components observed in K' are

$$\begin{cases} \vec{u}_1 = (u_{1x}\,,\, u_{1y}) = (\ \ u\,,\, 0) \\ \vec{u}_2 = (u_{2x}\,,\, u_{2y}) = (-u\,,\, 0) \end{cases} \qquad \text{and} \qquad \begin{cases} \vec{u}'_1 = (u'_{1x}\,,\, u'_{1y}) = (-u\cos\theta\,,\ \ u\sin\theta) \\ \vec{u}'_2 = (u'_{2x}\,,\, u'_{2y}) = (\ \ u\cos\theta\,,\, -u\sin\theta) \end{cases}$$

where $\theta = 0$ corresponds to the case of the headon collision with $\vec{u}'_1 = -\vec{u}_1$ and $\vec{u}'_2 = -\vec{u}_2$.

5(b) In Frame K where Particle 2 Is Rest Before the Collision

Now we consider this collision in reference frame K that is moving with respect to K' frame with velocity $\vec{u_2} = -\vec{u}$, where particle **2** is rest before the collision. Let $(\vec{v}_1\,,\,\vec{v}_2)$ and $(\vec{v}'_1\,,\,\vec{v}'_2)$ be the velocities of the particles in K frame before and after the collision, respectively,



$$\underset{(\vec{v}_1\,,\,\vec{v}_2)}{\text{before collision}} \qquad \xrightarrow{\text{collision}} \qquad \underset{(\vec{v}'_1\,,\,\vec{v}'_2)}{\text{after collision}}$$

Since K' frame is moving with respect to K frame with velocity

$$c\vec{\beta} = \vec{u} = u\,\vec{e}_x \qquad \text{and} \qquad \beta = u/c$$

the transformation of the particle velocities from K' to K can be directly obtained from Eq. (21). Before the collision

$$\begin{cases} v_{1x} = \dfrac{u + u_{1x}}{1 + \beta u_{1x}/c} = \dfrac{2c\beta}{1+\beta^2} \\ v_{1y} = 0 \end{cases} \qquad \text{and} \qquad \begin{cases} v_{2x} = \dfrac{u + u_{2x}}{1 + \beta u_{2x}/c} = 0 \\ v_{2y} = 0 \end{cases} \tag{49}$$

where $v_1 = v_{1x} = 2u/(1+\beta^2)$ and $v_2 = 0$. After the collision

$$\begin{cases} v'_{1x} = \dfrac{u + u'_{1x}}{1 + \beta u'_{1x}/c} = \dfrac{c\beta(1-\cos\theta)}{1-\beta^2\cos\theta} \overset{\theta=0}{=} 0 \\ v'_{1y} = \dfrac{u'_{1y}}{\gamma(1 + \beta u'_{1x}/c)} = \dfrac{c\beta\sin\theta}{\gamma(1-\beta^2\cos\theta)} \overset{\theta=0}{=} 0 \end{cases} \tag{50}$$

$$\begin{cases} v'_{2x} = \dfrac{u + u'_{2x}}{1 + \beta u'_{2x}/c} = \dfrac{c\beta(1+\cos\theta)}{1+\beta^2\cos\theta} \overset{\theta=0}{=} \dfrac{2c\beta}{1+\beta^2} \\ v'_{2y} = \dfrac{u'_{2y}}{\gamma(1 + \beta u'_{2x}/c)} = \dfrac{-c\beta\sin\theta}{\gamma(1+\beta^2\cos\theta)} \overset{\theta=0}{=} 0 \end{cases} \tag{51}$$

The conservation of momentum in the y direction observed in K frames is then

$$0 + 0 = \mathcal{M}(v'_1)\frac{c\beta\sin\theta}{\gamma(1-\beta^2\cos\theta)} - \mathcal{M}(v'_2)\frac{c\beta\sin\theta}{\gamma(1+\beta^2\cos\theta)}$$

which yields

$$\frac{\mathcal{M}(v_1')}{1-\beta^2\cos\theta} \;=\; \frac{\mathcal{M}(v_2')}{1+\beta^2\cos\theta}$$

At $\theta = 0$, $\vec{v}\,'_1 = 0$ and $\mathcal{M}(0) = m$,

$$\mathcal{M}(v_2') \;=\; \mathcal{M}(v_1')\frac{1+\beta^2\cos\theta}{1-\beta^2\cos\theta} \;=\; m\frac{1+\beta^2}{1-\beta^2}$$

Moreover, at $\theta = 0$,

$$\frac{v_2'}{c} \;=\; \frac{2\beta}{1+\beta^2} \qquad \text{and} \qquad 1-\left(\frac{v_2'}{c}\right)^2 \;=\; 1-\frac{4\beta^2}{(1+\beta^2)^2} \;=\; \frac{(1-\beta^2)^2}{(1+\beta^2)^2}$$

Therefore, at $\theta = 0$

$$\mathcal{M}(v_2') \;=\; \frac{m}{\sqrt{1-(v_2'/c)^2}} \;=\; \gamma_{v_2'}m$$

Since θ here is arbitrary and depends on choice of the direction of the coordinate axis while $\mathcal{M}$ is a scalar function and should be independent of the coordinate, in general, we can define

$$\mathcal{M}(u) \;=\; \gamma_u m$$

and the generalized momentum is defined as

$$\vec{p} \;=\; \mathcal{M}(u)\vec{u} \;=\; \gamma_u m\vec{u} \;=\; m\vec{U} \tag{52}$$

where $\vec{U}$ is the space components of the 4-velocity. The relationship between $\vec{p}$ and $\vec{U}$ in Eq. (52) suggests that the 4-vector for momentum should be defined through the 4-velocity as

$$(\,p_0\,,\,\vec{p}\,) = m(\,U_0\,,\,\vec{U}\,) = \gamma_u(\,mc\,,\,m\vec{u}\,) \tag{53}$$

where

$$p_0 = \gamma mc = \frac{mc}{\sqrt{1-u^2/c^2}} \;\xrightarrow{c\to\infty}\; \frac{1}{c}\left(mc^2+\frac{1}{2}mu^2\right)$$

The energy in Einstein relativity should thus be defined as

$$E = cp_0 = \gamma mc^2 \tag{54}$$

and $\mathbf{p} = (p_0, \vec{p})$ forms the energy-momentum 4-vector. The scalar product of $\mathbf{p}$,

$$\mathbf{p}^T\mathbf{g}\,\mathbf{p} = p_0^2 - \vec{p}\cdot\vec{p} = (mc)^2 \tag{55}$$

is invariant. The energy can also be written with the momentum as

$$E = \sqrt{c^2p^2+m^2c^4} \tag{56}$$

and at $p = 0$, $E = mc^2$ is the rest energy of a particle.

5(c) Verify the Energy Conservation During Elastic Collision

With the relativistic definition of the kinetic energy $E = \gamma mc^2$, the conservation of the energy for the elastic collision

$$E_1 + E_2 = E_{1'} + E_{2'} \qquad \Longrightarrow \qquad \gamma_1 + \gamma_2 = \gamma_{1'} + \gamma_{2'} \tag{57}$$

can be written in both CM frame K' and Lab frame K where particle **2** is statiojnary before the collision.

In Center-of-Mass Frame K'

The energy conservation is

$$\gamma_{u_1} + \gamma_{u_2} = \gamma_{u_1'} + \gamma_{u_2'} \tag{58}$$

Since $u_1 = u_2 = u_1' = u_2' = u$

$$\gamma_{u_1} = \gamma_{u_2} = \gamma_{u_1'} = \gamma_{u_2'} = \frac{1}{\sqrt{1 - u^2/c^2}}$$

The energy conservation in Eq. (58) is therefore true for the energy definition of $E = \gamma mc^2$.

In Frame K where Particle 2 Is Rest Before the Collision

The energy conservation is

$$\gamma_{v_1} + \gamma_{v_2} = \gamma_{v_1'} + \gamma_{v_2'} \tag{59}$$

From Eq. (49),

$$\gamma_{v_1} + \gamma_{v_2} = \frac{1+\beta^2}{1-\beta^2} + 1 = \frac{2}{1-\beta^2} \tag{60}$$

and from Eqs. (50) and (50)

$$\begin{aligned}
\frac{v_1'^2}{c^2} &= \frac{\beta^2}{(1+\beta^2\cos\theta)^2}\left[(1+\cos\theta)^2 + (1-\beta^2)\sin^2\theta\right] \\
\gamma_{v_1'}^{-2} &= 1 - \frac{v_1'^2}{c^2} \\
&= \frac{1}{(1+\beta^2\cos\theta)^2}\left[(1+\beta^2\cos\theta)^2 - \beta^2(1+\cos\theta)^2 - \beta^2(1-\beta^2)\sin^2\theta\right] \\
&= \frac{(1-\beta^2)^2}{(1+\beta^2\cos\theta)^2} \\
\frac{v_2'^2}{c^2} &= \frac{\beta^2}{(1-\beta^2\cos\theta)^2}\left[(1-\cos\theta)^2 + (1-\beta^2)\sin^2\theta\right] \\
\gamma_{v_2'}^{-2} &= \frac{1}{(1-\beta^2\cos\theta)^2}\left[(1-\beta^2\cos\theta)^2 - \beta^2(1-\cos\theta)^2 - \beta^2(1-\beta^2)\sin^2\theta\right] \\
&= \frac{(1-\beta^2)^2}{(1-\beta^2\cos\theta)^2}
\end{aligned}$$

Thus,

$$\gamma_{u_1'} + \gamma_{u_2'} = \frac{1+\beta^2\cos\theta}{1-\beta^2} + \frac{1-\beta^2\cos\theta}{1-\beta^2} = \frac{2}{1-\beta^2} \tag{61}$$

which equals Eq. (60) and the energy conservation in Eq. (59) is good.

6. Some Other Notations for Vector Calculations in Minkowski Space

To avoid using the Minkowski metric **g** in equations, two different forms of vectors, covariant and contravariant vector, are introduced for the vector calculations in the Minkowski space. For the coordinates, the two forms of the vector are

$$\text{Contravariant coordinate:} \qquad (x^0,\, x^1,\, x^2,\, x^3) \;=\; (\, x_0\,,\, x\,,\, y\,,\, z\,)$$
$$\text{Covariant coordinate:} \qquad (x_0,\, x_1,\, x_2,\, x_3) \;=\; (\, x_0\,,\, -x\,,\, -y\,,\, -z\,)$$

where the relatonship between the covariant and contravariant vector is

$$\begin{pmatrix} x_0 & x_0 & x_2 & x_3 \end{pmatrix}^T = \mathbf{g} \begin{pmatrix} x^0 & x^1 & x^2 & x^3 \end{pmatrix}^T \qquad \text{or} \qquad x_\alpha \;=\; \sum_{\beta=0}^{3} g_{\alpha\beta}\, x^\beta$$

where $g_{\alpha\beta}$ is the matrix elements of **g**. To simplify equations, the summation over an index can be written as a repeated index, *i.e.*

$$g_{\alpha\beta}\, x^\beta \quad \text{implies} \quad \sum_{\beta=0}^{3} g_{\alpha\beta}\, x^\beta$$

• For the covariant and contravariant vectors, the notation of the Lorentz transformation matrix **A** in Eq. (29) is

$$\frac{\partial x'^\alpha}{\partial x^\beta} = \mathbf{A} \qquad \text{and} \qquad \frac{\partial x^\beta}{\partial x'_\alpha} = \mathbf{A}^{-1} = \mathbf{g}\,\mathbf{A}^T\mathbf{g}$$

• The vector transformation between frames can be written in both contravariant and covariant form,

$$U'^\alpha = \frac{\partial x'^\alpha}{\partial x^\beta} U^\beta \qquad \text{and} \qquad U'_\alpha = \frac{\partial x^\beta}{\partial x'_\alpha} U_\beta$$

• The scalar invariant can be written as

$$\mathbf{U}\cdot\mathbf{V} = U_\alpha V^\alpha = U_\alpha g_{\alpha\beta} V_\beta = U_0 V_0 - \vec{U}\cdot\vec{V}$$

and

$$U'_\alpha V'^\alpha = \frac{\partial x^\beta}{\partial x'_\alpha} U_\beta \frac{\partial x'^\alpha}{\partial x_\eta} V^\eta = U_\beta\, \frac{\partial x^\beta}{\partial x_\eta}\, V^\eta = U_\beta\, \delta^\beta{}_\eta\, V^\eta = U_\beta V^\beta$$

• Transformation of Partial Derivative in Four-Dimensional Space

$$\frac{\partial F}{\partial x'^\alpha} = \frac{\partial x^\beta}{\partial x'^\alpha}\,\frac{\partial F}{\partial x^\beta} \qquad \Longrightarrow \qquad \frac{\partial}{\partial x'^\alpha} = \frac{\partial x^\beta}{\partial x'^\alpha}\,\frac{\partial}{\partial x^\beta}$$

The contravariant and covariant 4-dimensional divergence is defined as

$$\begin{aligned}\partial^\alpha &= \frac{\partial}{\partial x_\alpha} = \left(\frac{\partial}{\partial x_0}, \; -\nabla\right)\\ \partial_\alpha &= \frac{\partial}{\partial x^\alpha} = \left(\frac{\partial}{\partial x_0}, \; \nabla\right)\end{aligned}$$

and the divergence of any vector $\mathbf{U}$

$$\partial^\alpha U_\alpha = \partial_\alpha U^\alpha = \frac{\partial U^0}{\partial x_0} + \nabla \cdot \vec{U}$$

is invariant under the Lorentz transformation. The 4-dimensional Laplacian is defined as

$$\Box = \partial_\alpha \partial^\alpha = \frac{\partial^2}{\partial x_0 \partial x_0} - \nabla^2 = \vec{\partial}\,\mathbf{g}\,\vec{\partial}^T$$

and it is also invariant under the Lorentz transformation,

$$\vec{\partial}\,\mathbf{g}\,\vec{\partial}^T = \vec{\partial}\,\mathbf{A}^T\mathbf{g}\,\mathbf{A}\,\vec{\partial}^T = \vec{\partial'}\,\mathbf{g}\,\vec{\partial'}^T$$

where

$$\vec{\partial'} = \frac{\partial}{\partial x'^\alpha} = \left(\frac{\partial}{\partial x'_0} \quad \frac{\partial}{\partial x'_1} \quad \frac{\partial}{\partial x'_2} \quad \frac{\partial}{\partial x'_3}\right), \qquad \vec{\partial'}^T = \mathbf{A}\,\vec{\partial}^T \qquad \text{and} \qquad \vec{\partial'} = \vec{\partial}\,\mathbf{A}^T$$

Note that the Lorentz transformation is independent of $(t, \vec{r})$.

7. Infinitesimal Generators of Transformations in Minkowski Space

7.1. Number of Independent Parameters In Lorentz Group

All the Lorentz transformations, the Lorentz transformation matrix, form a group — Lorentz group, that is a subgroup of the symplectic group. The question here is how many independent parameters we need in order to parameterize the Lorentz group. A symplectic transformation in four-dimensional space satisfies

$$\mathbf{A}^T\mathbf{g}\mathbf{A} = \mathbf{g} \tag{62}$$

which in general yields $4 \times 4 = 16$ equations for 16 elements of $\mathbf{A}$. Since $\mathbf{g}$ is a symmetric matrix, $\mathbf{g}^T = \mathbf{g}$ and

$$\left(\mathbf{A}^T\mathbf{g}\mathbf{A}\right)^T = \mathbf{A}^T\mathbf{g}\mathbf{A}\,,$$

out of 16 equations in Eq. (62), the equations from the upper off-diagonal elements of $\mathbf{A}^T\mathbf{g}\mathbf{A}$ are the same as those from the lower off-diagonal elements of the matrix, *i.e.*

$$\left(\mathbf{A}^T\mathbf{g}\mathbf{A}\right)_{ij} = \mathbf{g}_{ij} \qquad \text{and} \qquad \left(\mathbf{A}^T\mathbf{g}\mathbf{A}\right)_{ji} = \mathbf{g}_{ji}$$

are the same equation. Therefore, the number of the independent equations in Eq. (62) is the number of diagonal elements plus the half of the off-diagonal elements. For the 4×4 matrix, the number of the independent equations of Eq. (62) is $4+3+2+1 = 10$. Therefore, there are $16 - 10 = 6$ independent parameters to specify all the elements in a Lorentz transformation matrix $\mathbf{A}$, *i.e.* the Lorentz group is a six-parameters symplectic group.

7.2. Lie Transformation for Lorentz Transformations

For a Lorentz transformation $\mathbf{A}$ between two inertial reference frames that move with a velocity $\vec{v} = c\vec{\beta}$ with respect to each other, since

$$\lim_{\beta \to 0} \mathbf{A} = \mathbf{I} \tag{63}$$

the Lorentz transformation is a continuous transformation and, moreover, the Lorentz is symplectic $\mathbf{A}^T\mathbf{g}\mathbf{A} = \mathbf{g}$. It can be written into a Lie transformation that is the transformations in the form of exponential function of an operator,

$$\mathbf{A} \;=\; e^{\mathbf{L}} \;=\; \sum_{n=0}^{\infty} \frac{1}{n!}\mathbf{L}^n \qquad \text{and} \qquad \mathbf{A}^{-1} \;=\; e^{-\mathbf{L}} \;=\; \sum_{n=0}^{\infty} \frac{(-1)^n}{n!}\mathbf{L}^n \tag{64}$$

where $\mathbf{L}$ is a 4×4 matrix. The presentation of $\mathbf{A}$ as a Lie transformation has the following properties.

7.2(a) If $\mathbf{U}^{-1}\mathbf{L}\mathbf{U} = \mathbf{\Lambda}$, where $\mathbf{\Lambda}$ is a diagonalized matrix, $\mathbf{U}$ also diagonalize $\mathbf{A}$ as

$$\mathbf{U}^{-1}\mathbf{A}\mathbf{U} \;=\; \mathbf{U}^{-1}e^{\mathbf{L}}\mathbf{U} \;=\; \sum_{n=0}^{\infty} \frac{1}{n!}\mathbf{U}^{-1}\mathbf{L}^n\mathbf{U} \;=\; \sum_{n=0}^{\infty} \frac{1}{n!}(\mathbf{U}^{-1}\mathbf{L}\mathbf{U})^n \;=\; e^{\mathbf{\Lambda}} \tag{65}$$

The eigenvalues of $\mathbf{A}$ is thus $\left(e^{\lambda_0}, e^{\lambda_1}, e^{\lambda_2}, e^{\lambda_3}\right)$ where $(\lambda_0, \lambda_1, \lambda_2, \lambda_3)$ are the eigenvalues of $\mathbf{L}$.

7.2(b) $\mathbf{L}$ is a traceless matrix. Since

$$1 \;=\; \det \mathbf{A} = \det(e^{\mathbf{\Lambda}}) \;=\; \prod_{i=0}^{3} e^{\lambda_i} \;=\; \exp\left(\sum_{i=0}^{3}\lambda_i\right) \;=\; e^{Tr\mathbf{L}}\,,$$

$$Tr\mathbf{L} = \sum_{i=0}^{3}\lambda_i = 0 \tag{66}$$

7.2(c) $\mathbf{gL}$ is an antisymmetric matrix. Since

$$\mathbf{A}^T \;=\; \sum_{n=0}^{\infty} \frac{1}{n!}\left(\mathbf{L}^T\right)^n \;=\; e^{\mathbf{L}^T} \qquad \Longrightarrow \qquad \mathbf{g}\,\mathbf{A}^T\mathbf{g} \;=\; \sum_{n=0}^{\infty} \frac{1}{n!}\left(\mathbf{g}\,\mathbf{L}^T\mathbf{g}\right)^n \;=\; e^{\mathbf{g}\,\mathbf{L}^T\mathbf{g}}$$

and

$$\mathbf{A}^T\mathbf{g}\mathbf{A} \;=\; \mathbf{g} \qquad \Longrightarrow \qquad \mathbf{g}\,\mathbf{A}^T\mathbf{g}\mathbf{A} \;=\; \mathbf{g}\mathbf{g} \;=\; \mathbf{I}$$

$\mathbf{g}\mathbf{A}^T\mathbf{g}$ is the inverse of $\mathbf{A}$,

$$\mathbf{A}^{-1} \;=\; e^{-\mathbf{L}} \;=\; \mathbf{g}\,\mathbf{A}^T\mathbf{g} \;=\; e^{\mathbf{g}\,\mathbf{L}^T\mathbf{g}}$$

Therefore

$$\mathbf{g}\,\mathbf{L}^T\mathbf{g} \;=\; -\mathbf{L} \qquad \Longrightarrow \qquad \mathbf{L}^T\mathbf{g} \;=\; (\mathbf{g}\,\mathbf{L})^T \;=\; -\mathbf{g}\,\mathbf{L} \tag{67}$$

i.e. $\mathbf{gL}$ is an antisymmetric matrix with

$$(\mathbf{gL})_{ij} = -(\mathbf{gL})_{ji} \qquad \text{and} \qquad (\mathbf{gL})_{ii} = 0 \tag{68}$$

7.2(d) The symmetry of matrix $\mathbf{L}$

Now let us take a close look of each matrix element of $\mathbf{gL}$,

$$(\mathbf{gL})_{ij} = \sum_{k=0}^{3} g_{ik} L_{kj} = \sum_{k=0}^{3} g_{ii}\delta_{ik} L_{kj} = g_{ii} L_{ij} \tag{69}$$

where $g_{ii} = 1$ for $i = 0$ and $g_{ii} = -1$ for $i = 1, 2, 3$. With Eqs. (68) and (69), therefore,

$$\begin{aligned} (\mathbf{gL})_{ii} = g_{ii} L_{ii} = 0 \qquad &\Longrightarrow \qquad L_{ii} = 0 \\ (\mathbf{gL})_{ij} = -(\mathbf{gL})_{ji} \qquad &\Longrightarrow \qquad g_{ii} L_{ij} = -g_{jj} L_{ji} \end{aligned} \tag{70}$$

For $i \neq j$, Eq. (70) requires that the off-diagonal matrix elements of $\mathbf{L}$ satisfies

$$\begin{cases} \text{for } i = 0 \text{ and } j = 1, 2, 3, \qquad L_{0j} = L_{j0}\,, \quad \text{Three independent element are } L_{01},\ L_{02},\ L_{03} \\ \text{for } i, j = 1, 2, 3, \qquad L_{ij} = -L_{ji}\,, \quad \text{Three independent element are } L_{12},\ L_{13},\ L_{23} \end{cases}$$

Recall that $\mathbf{A} = \exp(\mathbf{L})$ has six independent parameters for its 16 matrix elements. The matrix $\mathbf{L}$ with six independent elements is thus

$$\mathbf{L} = \begin{pmatrix} 0 & L_{01} & L_{02} & L_{03} \\ L_{01} & 0 & L_{12} & L_{13} \\ L_{02} & -L_{12} & 0 & L_{23} \\ L_{03} & -L_{13} & -L_{23} & 0 \end{pmatrix} \tag{71}$$

7.3. Infinitesimal Generators

The matrix $\mathbf{L}$ in Eq. (71) can be written as a sum of six base matrices that are six infinitesimal generators of the Lorentz group. Let

$$\begin{cases} \vec{\omega} = (\omega_1\,,\, \omega_2\,,\, \omega_3) = -(L_{23}\,,\, L_{13}\,,\, L_{12}) \\ \vec{\zeta} = (\zeta_1\,,\, \zeta_2\,,\, \zeta_3) = -(L_{01}\,,\, L_{02}\,,\, L_{03}) \end{cases} \tag{72}$$

and the six base matrices are defined as

$$\mathbf{S}_1 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \qquad \mathbf{S}_2 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \qquad \mathbf{S}_3 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\mathbf{K}_1 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \qquad \mathbf{K}_2 = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \qquad \mathbf{K}_3 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}$$

Then the matrix $\mathbf{L}$ and the transformation $\mathbf{A} = \exp(\mathbf{L})$ can be written as

$$\begin{cases} \mathbf{L} \;=\; -\vec{\omega}\cdot\vec{\mathbf{S}} - \vec{\zeta}\cdot\vec{\mathbf{K}} \\ \mathbf{A} \;=\; e^{\mathbf{L}} \;=\; e^{-\vec{\omega}\cdot\vec{\mathbf{S}}-\vec{\zeta}\cdot\vec{\mathbf{K}}} \end{cases} \tag{73}$$

where $\vec{\omega}$ are the three parameters for general rotational transformations in the Euclidian space and $\vec{\zeta}$ are the three parameters for the Lorentz"boosters" between two inertial frames moving in a arbitrary direction with respect to each other. It should be noted that the Lie transformation for $\mathbf{A}$ with six infinitesimal generators cannot be decomposed into a multiplication of Lie transformations with a single generator, *i.e.*

$$e^{-(\omega_1\mathbf{S}_1+\omega_2\mathbf{S}_2+\omega_3\mathbf{S}_3+\zeta_1\mathbf{K}_1+\zeta_2\mathbf{K}_2+\zeta_3\mathbf{K}_3)} \neq e^{-\omega_1\mathbf{S}_1}e^{-\omega_2\mathbf{S}_2}e^{-\omega_3\mathbf{S}_3}e^{-\zeta_1\mathbf{K}_1}e^{-\zeta_2\mathbf{K}_2}e^{-\zeta_3\mathbf{K}_3}$$

since the six base matrices are non-commuting (this topic will be discussed in the next section.) The commutators among the base matrices are

$$[\mathbf{S}_i,\, \mathbf{S}_j] = \epsilon_{ijk}\mathbf{S}_k\,, \qquad [\mathbf{S}_i,\, \mathbf{K}_j] = \epsilon_{ijk}\mathbf{K}_k\,, \qquad [\mathbf{K}_i,\, \mathbf{K}_j] = -\epsilon_{ijk}\mathbf{S}_k$$

Example 7.1. $\vec{\omega} = (0,\, 0,\, \omega)$ and $\vec{\zeta} = 0$

$$\mathbf{L} = -\omega\,\mathbf{S}_3 \qquad \text{and} \qquad \mathbf{A} = e^{-\omega\,\mathbf{S}_3} = \sum_{n=0}^{\infty} \frac{(-\omega)^n}{n!}\,\mathbf{S}_3^n \tag{74}$$

Since

$$\mathbf{S}_3^2 = -\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \qquad \text{and} \qquad \begin{cases} \mathbf{S}_3^{2m} & = & (\mathbf{S}_3^2)^m & = & (-1)^m\,(-\mathbf{S}_3^2) \\ \mathbf{S}_3^{2m-1} & = & \mathbf{S}_3^{2(m-1)}\,\mathbf{S}_3 & = & (-1)^m\,(-\mathbf{S}_3) \end{cases}$$

where $\mathbf{S}_3^2\,\mathbf{S}_3 = -\mathbf{S}_3\,,$

$$\begin{aligned} \mathbf{A} \;=\; \sum_{n=0}^{\infty} \frac{(-\omega)^n}{n!}\,\mathbf{S}_3^n \;&=\; \mathbf{I} + \sum_{m=1}^{\infty}\left[\frac{(-\omega)^{2m}}{(2m)!}\,\mathbf{S}_3^{2m} + \frac{(-\omega)^{2m-1}}{(2m-1)!}\,\mathbf{S}_3^{2m-1}\right] \\ &=\; \mathbf{I} + \left(-\mathbf{S}_3^2\right)\sum_{m=1}^{\infty}\frac{(-1)^m\omega^{2m}}{(2m)!} + (-\mathbf{S}_3)\sum_{m=1}^{\infty}\frac{(-1)^m\omega^{2m-1}}{(2m-1)!} \\ &=\; \mathbf{I} + \left(-\mathbf{S}_3^2\right)(\cos\omega - 1) + (-\mathbf{S}_3)\sin\omega \\ &=\; \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\omega & \sin\omega & 0 \\ 0 & -\sin\omega & \cos\omega & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \end{aligned}$$

This is a rotational transformation in the x-y plane with ω as the rotational angle.

Example 7.2. $\vec{\omega} = 0$ and $\vec{\zeta} = (\zeta,\, 0,\, 0)$

$$\mathbf{L} = -\zeta\, \mathbf{K}_1 \qquad \text{and} \qquad \mathbf{A} = e^{-\zeta\, \mathbf{K}_1} = \sum_{n=0}^{\infty} \frac{(-\zeta)^n}{n!}\, \mathbf{K}_1^n \tag{75}$$

Since

$$\mathbf{K}_1^2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \qquad \text{and} \qquad \begin{cases} \mathbf{K}_1^{2m} & = \ \mathbf{K}_1^2 \\ \mathbf{K}_1^{2m-1} & = \ \mathbf{K}_1^{2(m-1)}\, \mathbf{K}_1 \ = \ \mathbf{K}_1 \end{cases},$$

$$\begin{aligned}
\mathbf{A} \ &= \ \sum_{n=0}^{\infty} \frac{(-\zeta)^n}{n!}\, \mathbf{K}_1^n \ = \ \mathbf{I} + \sum_{m=1}^{\infty} \left[\frac{(-\zeta)^{2m}}{(2m)!}\, \mathbf{K}_1^{2m} + \frac{(-\zeta)^{2m-1}}{(2m-1)!}\, \mathbf{K}_1^{2m-1} \right] \\
&= \ \mathbf{I} \, + \, \mathbf{K}_1^2 \sum_{m=1}^{\infty} \frac{\zeta^{2m}}{(2m)!} - \mathbf{K}_1 \sum_{m=1}^{\infty} \frac{\zeta^{2m-1}}{(2m-1)!} \\
&= \ \mathbf{I} \, + \, \mathbf{K}_1^2\, (\cosh\zeta - 1) \, - \, \mathbf{K}_1 \sinh\zeta \\
&= \ \begin{pmatrix} \cosh\zeta & -\sinh\zeta & 0 & 0 \\ -\sinh\zeta & \cosh\zeta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \ \xrightarrow{\text{want to be}} \ \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}
\end{aligned}$$

where the last matrix is the Lorentz transformation with $\vec{v} \parallel \vec{e}_1$, which is what we want for $\mathbf{A}$. If $\cosh\zeta = \gamma$ and $\sinh\zeta = \gamma\beta$ with

$$\cosh^2\zeta - \sinh^2\zeta = \gamma^2 - \gamma^2\beta^2 = 1\,, \qquad \tanh\zeta = \beta\,, \qquad \zeta = \tanh^{-1}\beta$$

the transformation $\mathbf{A} = e^{-\zeta\, \mathbf{K}_1}$ is therefore the Lorentz transformation with $\vec{v} \parallel \vec{e}_1$.

Example 7.3. A Lorentz transformation with $\vec{v}$ in an arbitrary direction without rotation,

$$\vec{\omega} = 0 \qquad \text{and} \qquad \mathbf{L} \ = \ -\zeta_1 \mathbf{K}_1 - \zeta_2 \mathbf{K}_2 - \zeta_3 \mathbf{K}_3$$

To calculate $\mathbf{A} = e^{\mathbf{L}}$, we need to calculate $\mathbf{L}^n$. Let

$$\boldsymbol{\epsilon} = \frac{1}{\zeta} \begin{pmatrix} \zeta_1 & \zeta_2 & \zeta_3 \end{pmatrix} \qquad \text{with} \qquad \boldsymbol{\epsilon}\boldsymbol{\epsilon}^T = 1$$

be a row vector along the direction of $\vec{\zeta}$, where $\zeta^2 = \zeta_1^2 + \zeta_2^2 + \zeta_3^2$ and $\boldsymbol{\epsilon}^T$ is the column vector of $\boldsymbol{\epsilon}$. Then,

$$\mathbf{L} \;=\; \begin{pmatrix} 0 & -\zeta_1 & -\zeta_2 & -\zeta_3 \\ -\zeta_1 & 0 & 0 & 0 \\ -\zeta_2 & 0 & 0 & 0 \\ -\zeta_3 & 0 & 0 & 0 \end{pmatrix} \;=\; -\zeta\,\mathbf{M} \qquad \text{where} \qquad \mathbf{M} \;=\; \begin{pmatrix} 0 & \boldsymbol{\epsilon} \\ \boldsymbol{\epsilon}^T & \hat{\mathbf{0}}_3 \end{pmatrix}$$

and $\hat{\mathbf{0}}_3$ is the 3×3 zero matrix.

$$\mathbf{M}^2 \;=\; \begin{pmatrix} 0 & \boldsymbol{\epsilon} \\ \boldsymbol{\epsilon}^T & \hat{\mathbf{0}}_3 \end{pmatrix}\begin{pmatrix} 0 & \boldsymbol{\epsilon} \\ \boldsymbol{\epsilon}^T & \hat{\mathbf{0}}_3 \end{pmatrix} \;=\; \begin{pmatrix} 1 & \mathbf{0} \\ \mathbf{0}^T & \boldsymbol{\epsilon}^T\boldsymbol{\epsilon} \end{pmatrix}$$

where $\boldsymbol{\epsilon}^T\boldsymbol{\epsilon}$ is a 3×3 matrix with matrix elements $\left(\boldsymbol{\epsilon}^T\boldsymbol{\epsilon}\right)_{ij} = \zeta_i\zeta_j/\zeta^2$ and $\mathbf{0} = \begin{pmatrix} 0 & 0 & 0 \end{pmatrix}$.

$$\mathbf{M}^3 \;=\; \mathbf{M}\,\mathbf{M}^2 \;=\; \begin{pmatrix} 0 & \boldsymbol{\epsilon} \\ \boldsymbol{\epsilon}^T & \hat{\mathbf{0}}_3 \end{pmatrix}\begin{pmatrix} 1 & \mathbf{0} \\ \mathbf{0}^T & \boldsymbol{\epsilon}^T\boldsymbol{\epsilon} \end{pmatrix} \;=\; \begin{pmatrix} 0 & \boldsymbol{\epsilon} \\ \boldsymbol{\epsilon}^T & \hat{\mathbf{0}}_3 \end{pmatrix} \;=\; \mathbf{M}$$

where $\boldsymbol{\epsilon}\boldsymbol{\epsilon}^T\boldsymbol{\epsilon} = \boldsymbol{\epsilon}$. Therefore,

$$\left.\begin{array}{lllll} \mathbf{M}^3 & = & \mathbf{M}\,\mathbf{M}^2 & = & \mathbf{M} \\ \mathbf{M}^4 & = & \mathbf{M}\,\mathbf{M}^3 & = & \mathbf{M}^2 \\ \mathbf{M}^5 & = & \mathbf{M}\,\mathbf{M}^4 & = & \mathbf{M}^3 \;=\; \mathbf{M} \\ \mathbf{M}^6 & = & \mathbf{M}\,\mathbf{M}^5 & = & \mathbf{M}^2 \\ \mathbf{M}^7 & = & \mathbf{M}\,\mathbf{M}^6 & = & \mathbf{M}^3 \;=\; \mathbf{M} \\ \mathbf{M}^8 & = & \mathbf{M}\,\mathbf{M}^7 & = & \mathbf{M}^2 \\ \vdots & & & & \vdots \end{array}\right\} \quad\Longrightarrow\quad \left\{\begin{array}{lll} \mathbf{M}^{2n} & = & \mathbf{M}^2 \\ \mathbf{M}^{2n-1} & = & \mathbf{M} \end{array}\right. \quad \text{where } n = 1, 2, 3, \ldots.$$

Therefore,

$$\begin{aligned} \mathbf{A} \;=\; e^{-\zeta\mathbf{M}} \;&=\; \sum_{n=0}^{\infty} \frac{(-\zeta)^n}{n!}\,\mathbf{M}^n \\ &=\; \mathbf{I} + \sum_{n=1}^{\infty}\left[\frac{(-\zeta)^{(2n)}}{(2n)!}\,\mathbf{M}^{2n} + \frac{(-\zeta)^{(2n-1)}}{(2n-1)!}\,\mathbf{M}^{2n-1}\right] \\ &=\; \mathbf{I} + \mathbf{M}^2\sum_{n=1}^{\infty}\frac{\zeta^{(2n)}}{(2n)!} - \mathbf{M}\sum_{n=1}^{\infty}\frac{\zeta^{(2n-1)}}{(2n-1)!} \\ &=\; \mathbf{I} + \mathbf{M}^2\left(\cosh\zeta - 1\right) - \mathbf{M}\sinh\zeta \end{aligned} \tag{76}$$

where

$$\mathbf{M} = \frac{1}{\zeta}\begin{pmatrix} 0 & \zeta_1 & \zeta_2 & \zeta_3 \\ \zeta_1 & 0 & 0 & 0 \\ \zeta_2 & 0 & 0 & 0 \\ \zeta_3 & 0 & 0 & 0 \end{pmatrix} \qquad \text{and} \qquad \mathbf{M}^2 = \frac{1}{\zeta^2}\begin{pmatrix} \zeta^2 & 0 & 0 & 0 \\ 0 & \zeta_1^2 & \zeta_1\zeta_2 & \zeta_1\zeta_3 \\ 0 & \zeta_1\zeta_2 & \zeta_2^2 & \zeta_2\zeta_3 \\ 0 & \zeta_1\zeta_3 & \zeta_2\zeta_3 & \zeta_3^2 \end{pmatrix}$$

To write matrix $\mathbf{A}$ in Eq. (76) into the explicit form of the Lorentz transformation in arbitrary direction in Eq. (29), let

$$\frac{1}{\zeta}\,(\zeta_1\,,\,\zeta_2\,,\,\zeta_3) = \frac{1}{\beta}\,(\beta_x\,,\,\beta_y\,,\,\beta_z) \qquad \Longrightarrow \qquad \frac{\zeta_i\zeta_j}{\zeta^2} = \frac{\beta_i\beta_j}{\beta^2} \tag{77}$$

where $\beta^2 = \beta_x^2 + \beta_y^2 + \beta_z^2$ and $\zeta^2 = \zeta_1^2 + \zeta_2^2 + \zeta_3^2$. Note that this problem starts with $\vec{\zeta} = (\zeta_1, \zeta_2, \zeta_3)$ as an arbitrary vector with three arbitrary components. The relationship between $\vec{\zeta}$ and $\vec{\beta}$ in Eq. (77) only aligns the direction of $\vec{\zeta}$ with the direction of $\vec{\beta}$, while the magnitude of $\vec{\zeta}$ is still arbitrary and undetermined. We can therefore choose the magnitude of $\vec{\zeta}$, similar to Example 2, as

$$\zeta = \tanh^{-1}\beta\,, \qquad \cosh\zeta = \gamma\,, \qquad \text{and} \qquad \sinh\zeta = \gamma\beta$$

where $\cosh^2\zeta - \sinh^2\zeta = \gamma^2 - \gamma^2\beta^2 = 1$. Then the transformation $\mathbf{A}$ in Eq. (76) becomes

$$\mathbf{A} \;=\; e^{-\vec{\zeta}\cdot\vec{\mathbf{K}}} \;=\; \begin{pmatrix} \gamma & -\gamma\beta_x & -\gamma\beta_y & -\gamma\beta_z \\ -\gamma\beta_x & 1+\dfrac{(\gamma-1)\beta_x^2}{\beta^2} & \dfrac{(\gamma-1)\beta_x\beta_y}{\beta^2} & \dfrac{(\gamma-1)\beta_x\beta_z}{\beta^2} \\ -\gamma\beta_y & \dfrac{(\gamma-1)\beta_x\beta_y}{\beta^2} & 1+\dfrac{(\gamma-1)\beta_y^2}{\beta^2} & \dfrac{(\gamma-1)\beta_y\beta_z}{\beta^2} \\ -\gamma\beta_z & \dfrac{(\gamma-1)\beta_x\beta_z}{\beta^2} & \dfrac{(\gamma-1)\beta_y\beta_z}{\beta^2} & 1+\dfrac{(\gamma-1)\beta_z^2}{\beta^2} \end{pmatrix}$$

which is the general Lorentz transformation with the velocity in the direction of $\vec{\beta}/\beta$.

8. Lorentz Group is an Non-Commuting Group

In general, the order of Lie transformations cannot be altered,

$$\mathbf{A_2A_1} \;=\; e^{\mathbf{L}_2}e^{\mathbf{L}_1} \;\neq\; e^{\mathbf{L}_1+\mathbf{L}_2} \;\neq\; e^{\mathbf{L}_1}e^{\mathbf{L}_2} \;=\; \mathbf{A_1A_2}$$

Two Lie transformations could be combined into a single one using Baker-Campbell-Hausdorff (BCH) formula

$$\mathbf{A_2A_1} = e^{\mathbf{L}_2}e^{\mathbf{L}_1} = e^{\mathbf{L}_1+\mathbf{L}_2+\mathbf{C}}$$

$$\mathbf{C} = \frac{1}{2!}\,[\mathbf{L}_2,\,\mathbf{L}_1] + \frac{1}{3!}\,[\mathbf{L}_2,\,[\mathbf{L_2},\,\mathbf{L}_1]] + \frac{1}{3!}\,[\mathbf{L}_1,\,[\mathbf{L_1},\,\mathbf{L}_2]] + \frac{1}{4!}\,[\mathbf{L}_1,\,[\mathbf{L}_1,\,[\mathbf{L_1},\,\mathbf{L}_2]]] + \cdots$$

Since six infinitesimal generators $(\mathbf{S}_1, \mathbf{S}_2, \mathbf{S}_3, \mathbf{K}_1, \mathbf{K}_3, \mathbf{K}_3)$ of the Lorentz group are non-commuting,

$$[\mathbf{S}_i\,,\,\mathbf{S}_j] \neq 0\,, \qquad [\mathbf{S}_i\,,\,\mathbf{K}_j] \neq 0\,, \qquad [\mathbf{K}_i\,,\,\mathbf{K}_j] \neq 0\,, \qquad \text{for } i \neq j,$$

two successive Lorentz transformations is in general not the same as a single transformation with a combined velocity.

Example 8.1. If two Lorentz transformations are in the same direction, they are commute

$$\mathbf{A}_i = \exp\left(-\zeta_i \mathbf{K}_1\right) = \begin{pmatrix} \cosh\zeta_i & -\sinh\zeta_i & 0 & 0 \\ -\sinh\zeta_i & \cosh\zeta_i & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} \gamma_i & -\gamma_i\beta_i & 0 & 0 \\ -\gamma_i\beta_i & \gamma_i & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

where $i = 1, 2$ and

$$\cosh\zeta_i = \gamma_i\,, \qquad \sinh\zeta_i = \gamma_i\beta_i$$

Since $[\mathbf{K}_1\,, \mathbf{K}_1] = 0$,

$$\mathbf{A}_2\mathbf{A}_1 \;=\; e^{-\zeta_2\mathbf{K}_1}e^{-\zeta_1\mathbf{K}_1} \;=\; e^{-(\zeta_2+\zeta_1)\mathbf{K}_1} \;=\; e^{-\zeta_1\mathbf{K}_1}e^{-\zeta_2\mathbf{K}_1} \;=\; \mathbf{A}_1\mathbf{A}_2$$

The combined transformation can be easily obtained as

$$\begin{pmatrix} \cosh\zeta_1 & -\sinh\zeta_1 \\ -\sinh\zeta_1 & \cosh\zeta_1 \end{pmatrix} \begin{pmatrix} \cosh\zeta_2 & -\sinh\zeta_2 \\ -\sinh\zeta_2 & \cosh\zeta_2 \end{pmatrix} = \begin{pmatrix} \cosh(\zeta_1+\zeta_2) & -\sinh(\zeta_1+\zeta_2) \\ -\sinh(\zeta_1+\zeta_2) & \cosh(\zeta_1+\zeta_2) \end{pmatrix}$$

where

$$\begin{aligned} \gamma &= \cosh(\zeta_1+\zeta_2) = \cosh\zeta_1\cosh\zeta_2 + \sinh\zeta_1\sinh\zeta_2 = \gamma_1\gamma_2(1+\beta_1\beta_2) \\ \gamma\beta &= \sinh(\zeta_1+\zeta_2) = \cosh\zeta_1\sinh\zeta_2 + \cosh\zeta_1\sinh\zeta_2 = \gamma_1\gamma_2(\beta_1+\beta_2) \end{aligned}$$

and

$$\mathbf{A}_1\mathbf{A_2} = e^{-(\zeta_1+\zeta_2)\mathbf{K}_1} = \begin{pmatrix} \cosh(\zeta_1+\zeta_2) & -\sinh(\zeta_1+\zeta_2) & 0 & 0 \\ -\sinh(\zeta_1+\zeta_2) & \cosh(\zeta_1+\zeta_2) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

where

$$\gamma = \gamma_1\gamma_2(1+\beta_1\beta_2)\,, \qquad \beta = \frac{\beta_1+\beta_2}{1+\beta_1\beta_2} \qquad \text{and} \qquad \gamma = 1/\sqrt{1-\beta^2}$$

which satisfies the addition of two velocities that are along the same direction.

Example 8.2. If two Lorentz transformations are in different directions, they are not commute. For example,

$$\mathbf{A}_1 = e^{-\zeta_1\mathbf{K}_1} = \begin{pmatrix} \cosh\zeta_1 & -\sinh\zeta_1 & 0 & 0 \\ -\sinh\zeta_1 & \cosh\zeta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} \gamma_1 & -\gamma_1\beta_1 & 0 & 0 \\ -\gamma_1\beta_1 & \gamma_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\mathbf{A}_2 = e^{-\zeta_2\mathbf{K}_2} = \begin{pmatrix} \cosh\zeta_2 & 0 & -\sinh\zeta_2 & 0 \\ 0 & 1 & 0 & 0 \\ -\sinh\zeta_2 & 0 & \cosh\zeta_2 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} \gamma_2 & 0 & -\gamma_2\beta_2 & 0 \\ 0 & 1 & 0 & 0 \\ -\gamma_2\beta_2 & 0 & \gamma_2 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\mathbf{A}_1\mathbf{A}_2 = e^{-(\zeta_1\mathbf{K}_1+\zeta_2\mathbf{K}_2)+\mathbf{C}} \;\neq\; \mathbf{A}_2\mathbf{A}_1$$

where

$$\mathbf{C} = \frac{\alpha_1}{2!}\,[\mathbf{K}_1,\,\mathbf{K}_2] + \frac{\alpha_{21}}{3!}\,[\mathbf{K}_1,\,[\mathbf{K}_1,\,\mathbf{K}_2]] + \frac{\alpha_{22}}{3!}\,[\mathbf{K}_2,\,[\mathbf{K}_2,\,\mathbf{K}_1]] + \cdots$$

and

$$\begin{aligned}
[\mathbf{K}_1,\,\mathbf{K}_2] &= \mathbf{S}_3 && \longleftarrow \text{rotation} \\
[\mathbf{K}_1,\,[\mathbf{K}_1,\,\mathbf{K}_2]] &= [\mathbf{K}_1,\,\mathbf{S}_3] \;=\; -\mathbf{K}_2 \\
[\mathbf{K}_2,\,[\mathbf{K}_2,\,\mathbf{K}_1]] &= -\,[\mathbf{K}_2,\,\mathbf{S}_3] \;=\; -\mathbf{K}_1 \\
&\;\vdots
\end{aligned}$$

Thus, $\mathbf{A}_1\mathbf{A}_2$ is a complicated combination of transformations of rotations and Lorentz boosters. The non-commuting of $\mathbf{A}_1$ and $\mathbf{A}_2$ can also be directly verified.

$$\mathbf{A}_1\mathbf{A}_2 = \begin{pmatrix} \gamma_1\gamma_2 & -\gamma_1\beta_1 & -\gamma_1\gamma_2\beta_2 & 0 \\ -\gamma_1\gamma_2\beta_1 & \gamma_1 & -\gamma_1\gamma_2\beta_1\beta_2 & 0 \\ -\gamma_2\beta_2 & 0 & \gamma_2 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Since $\mathbf{A}_1^T = \mathbf{A}_1$ and $\mathbf{A}_2^T = \mathbf{A}_2$,

$$(\mathbf{A}_1\mathbf{A}_2)^T \;=\; \mathbf{A}_2^T\mathbf{A}_1^T \;=\; \mathbf{A}_2\mathbf{A}_1$$

As $\mathbf{A}_1\mathbf{A}_2$ is not a symmetrical matrix,

$$\mathbf{A}_2\mathbf{A}_1 \;\neq\; \mathbf{A}_1\mathbf{A}_2$$

9. Review of Electrodynamics In Vacuum —— in Gaussian Unit

9(a) Macroscopic Maxwell Equations in Vacuum

$$\begin{aligned}
&\nabla\cdot\vec{E} = 4\pi\rho\,, && \nabla\times\vec{E} + \frac{1}{c}\frac{\partial\vec{B}}{\partial t} = 0 \\
&\nabla\cdot\vec{B} = 0 \quad , && \nabla\times\vec{B} - \frac{1}{c}\frac{\partial\vec{E}}{\partial t} = \frac{4\pi}{c}\vec{J}
\end{aligned} \tag{78}$$

9(b) Lorentz Force

$$\vec{F} = q\left(\vec{E} + \frac{1}{c}\vec{v}\times\vec{B}\right) \tag{79}$$

9(c) Continuity Equation

From the inhomogeneous (1st and 4th) Maxwell's equations

$$\frac{4\pi}{c}\nabla\cdot\vec{J} = \nabla\cdot\left(\nabla\times\vec{B} - \frac{1}{c}\frac{\partial\vec{E}}{\partial t}\right) = -\frac{1}{c}\frac{\partial}{\partial t}(\nabla\cdot\vec{E}) = -\frac{4\pi}{c}\frac{\partial\rho}{\partial t}$$

which yields the continuity equation of charges,

$$\frac{\partial\rho}{\partial t} + \nabla\cdot\vec{J} = 0 \tag{80}$$

9(d) Vector and Scalar Potentials

Because $\nabla\cdot\vec{B} = 0$, we can define

$$\vec{B} = \nabla\times\vec{A} \tag{81}$$

Then from another homogenous Maxwell's equation

$$\nabla\times\vec{E} + \frac{1}{c}\frac{\partial\vec{B}}{\partial t} = \nabla\times\left(\vec{E} + \frac{1}{c}\frac{\partial\vec{A}}{\partial t}\right) = 0$$

We can define

$$\vec{E} + \frac{1}{c}\frac{\partial\vec{A}}{\partial t} = -\nabla\Phi \qquad \text{or} \qquad \vec{E} = -\nabla\Phi - \frac{1}{c}\frac{\partial\vec{A}}{\partial t} \tag{82}$$

9(e) Wave Equations

Using the potentials in two inhomogeneous Maxwell equations yields

$$4\pi\rho = \nabla\cdot\vec{E} = \nabla\cdot\left(-\nabla\Phi - \frac{1}{c}\frac{\partial\vec{A}}{\partial t}\right) = -\nabla^2\Phi - \frac{1}{c}\frac{\partial}{\partial t}(\nabla\cdot\vec{A}) \tag{83}$$

and

$$\frac{4\pi}{c}\vec{J} = \nabla\times\vec{B} - \frac{1}{c}\frac{\partial\vec{E}}{\partial t} = \nabla\times(\nabla\times\vec{A}) + \frac{1}{c}\left[\nabla\left(\frac{\partial\Phi}{\partial t}\right) + \frac{1}{c}\frac{\partial^2\vec{A}}{\partial t^2}\right] \tag{84}$$

Since

$$\nabla\times(\nabla\times\vec{A}) = \nabla(\nabla\cdot\vec{A}) - \nabla^2\vec{A}$$

Eq. (84) can be written as

$$\frac{1}{c^2}\frac{\partial^2\vec{A}}{\partial t^2} - \nabla^2\vec{A} + \nabla\left(\nabla\cdot\vec{A} + \frac{1}{c}\frac{\partial\Phi}{\partial t}\right) = \frac{4\pi}{c}\vec{J} \tag{85}$$

Lorentz Gauge

To have a set of clean-look wave equations in Eq. (85), we need to choose a proper set of potentials $(\Phi,\,\vec{A})$ such that

$$\nabla\cdot\vec{A} + \frac{1}{c}\frac{\partial\Phi}{\partial t} = 0 \qquad \text{—— Lorentz Gauge}$$

Why can we do this? Consider a gauge transformation—a transformation of ϕ and $\vec{A}$ that keeps $\vec{B}$ and $\vec{E}$ unchanged, *i.e.*

$$(\Phi,\,\vec{A}) \qquad \longrightarrow \qquad (\Phi',\,\vec{A}\,')$$

$$\vec{B} = \nabla \times \vec{A} = \nabla \times \vec{A}' \qquad \Longrightarrow \qquad \vec{A}' = \vec{A} + \nabla f$$

where $f = f(\vec{r}, t)$ is an arbitrary function. Moreover

$$\vec{E} = -\nabla\Phi - \frac{1}{c}\frac{\partial \vec{A}}{\partial t} = -\nabla\Phi' - \frac{1}{c}\frac{\partial \vec{A}'}{\partial t} = -\nabla\Phi' - \frac{1}{c}\frac{\partial}{\partial t}\left(\vec{A} + \nabla f\right)$$

$$\Longrightarrow \qquad \Phi' = \Phi - \frac{1}{c}\frac{\partial f}{\partial t}$$

Therefore, we can always choose a set of $(\Phi', \vec{A}')$ or choose a proper function f such that

$$\nabla \cdot \vec{A}' + \frac{1}{c}\frac{\partial \Phi'}{\partial t} = \nabla \cdot \vec{A} + \frac{1}{c}\frac{\partial \Phi}{\partial t} + \left(\nabla^2 f - \frac{1}{c^2}\frac{\partial^2 f}{\partial t^2}\right) = 0$$

With the Lorentz gauge, the electromagnetic wave satisfies

$$\left\{ \begin{array}{rcl} \dfrac{1}{c^2}\dfrac{\partial^2 \vec{A}}{\partial t^2} - \nabla^2 \vec{A} & = & \dfrac{4\pi}{c}\vec{J} \\ \dfrac{1}{c^2}\dfrac{\partial^2 \Phi}{\partial t^2} - \nabla^2 \Phi & = & 4\pi\Phi \end{array} \right. \tag{86}$$

10. Covariant of Electrodynamics

10(a) Continuity Equation

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \vec{J} = 0 \tag{87}$$

Because the 4-vector for partial derivatives is

$$\partial_\alpha = \left(\frac{\partial}{\partial x_0}\,, \;\; \nabla \right)$$

The 4-vector for the charge/current can be chosen as

$$J^\alpha = \left(c\rho\,, \; \vec{J} \right) \tag{88}$$

The continuity equation becomes a scalar in the Minkowski space,

$$\partial_\alpha J^\alpha = 0 \tag{89}$$

and it is invariant under the Lorentz transformation.

10(b) Wave Equation

The 4-vector potential can be defined as

$$A^\alpha = (\Phi\,, \;\; \vec{A}) \tag{90}$$

and the 4-dimensional Laplacian is

$$\Box = \partial_\alpha \partial^\alpha = \partial_\alpha\, \mathbf{g}\, \partial_\alpha^T = \frac{\partial^2}{\partial x_0^2} - \nabla^2$$

The wave equations can be written into a 4-vector equation as

$$\left\{\begin{array}{rcl} \dfrac{1}{c^2}\dfrac{\partial^2\vec{A}}{\partial t^2} - \nabla^2\vec{A} &=& \dfrac{4\pi}{c}\vec{J} \\ \dfrac{1}{c^2}\dfrac{\partial^2\Phi}{\partial t^2} - \nabla^2\Phi &=& 4\pi\Phi \end{array}\right. \qquad \Longrightarrow \qquad \Box A^\alpha = \frac{4\pi}{c}J^\alpha \tag{91}$$

10(c) Equation of Motion in $\vec{E}$ and $\vec{B}$ Fields

The equation of motion of a charge particle in $\vec{E}$ and $\vec{B}$ fields is

$$\frac{d\vec{p}}{dt} = q\left(\vec{E} + \frac{1}{c}\vec{v}\times\vec{B}\right)$$

Since the 4-velocity and 4-momentum are defined as

$$\left\{\begin{array}{llll} U^\alpha = (U_0\,,\ \vec{U}) & \text{with} & U_0 = \dfrac{dx_0}{d\tau} = \gamma c & \text{and} \quad \vec{U} = \dfrac{d\vec{r}}{d\tau} = \gamma\vec{v} \\ p^\alpha = (p_0\,,\ \vec{p}) & \text{with} & p_0 = \dfrac{E}{c} = mU_0 & \text{and} \quad \vec{p} = m\vec{U} \end{array}\right.$$

For the 4-vector equation of motion, the three space-components can be constructed as

$$\gamma\frac{d\vec{p}}{dt} = q\left(\gamma\vec{E} + \frac{1}{c}\gamma\vec{v}\times\vec{B}\right) \qquad \Longrightarrow \qquad \frac{d\vec{p}}{d\tau} = \frac{q}{c}\left(U_0\vec{E} + \vec{U}\times\vec{B}\right)$$

where $dt = \gamma d\tau$. To construct the time-component of the equation of motion, we need $dp_0/d\tau$. Since

$$p_\alpha p^\alpha = p_0^2 - \vec{p}\cdot\vec{p} = (mc)^2 \qquad \Longrightarrow \qquad p_0^2 = (mc)^2 + \vec{p}\cdot\vec{p},$$

$$p_0\frac{dp_0}{d\tau} = \vec{p}\cdot\frac{d\vec{p}}{d\tau} = \frac{q}{c}\,\vec{p}\cdot\left(U_0\vec{E} + \vec{U}\times\vec{B}\right) = \frac{q}{c}U_0\,\vec{p}\cdot\vec{E}$$

where $\vec{p}\cdot(\vec{U}\times\vec{B}) = 0$ and $U_0\vec{p} = p_0\vec{U}$. Therefore,

$$\frac{dp_0}{d\tau} = \frac{q}{c}\,\vec{U}\cdot\vec{E}$$

The 4-vector form of the equation of motion can then be written as

$$\left\{\begin{array}{rcl} \dfrac{dp_0}{d\tau} &=& \dfrac{q}{c}\,\vec{U}\cdot\vec{E} \\ \dfrac{d\vec{p}}{d\tau} &=& \dfrac{q}{c}\left(U_0\vec{E} + \vec{U}\times\vec{B}\right) \end{array}\right. \qquad \Longrightarrow \qquad \left\{\begin{array}{rcl} \dfrac{c}{q}\dfrac{dp_0}{d\tau} &=& 0U_0 + E_1U_1 + E_2U_2 + E_3U_3 \\ \dfrac{c}{q}\dfrac{dp_1}{d\tau} &=& E_1U_0 + 0U_1 + B_3U_2 - B_2U_3 \\ \dfrac{c}{q}\dfrac{dp_2}{d\tau} &=& E_2U_0 - B_3U_1 + 0U_2 + B_1U_3 \\ \dfrac{c}{q}\dfrac{dp_3}{d\tau} &=& E_3U_0 + B_2U_1 - B_1U_2 + 0U_3 \end{array}\right.$$

which can be written into a 4-vector form

$$\frac{dp^\alpha}{d\tau} = \frac{q}{c} F^{\alpha\beta} g_{\beta\eta} U^\eta = \frac{q}{c} F^{\alpha\beta} U_\beta \tag{92}$$

where

$$U^\beta = (U_0,\, U_1,\, U_2,\, U_3)^T\,, \qquad\qquad U_\beta = g_{\beta\eta} U^\eta = (U_0,\, -U_1,\, -U_2,\, -U_3)^T$$

and $F^{\alpha\beta}$ is the field strength tensor,

$$\mathbf{F} \;=\; F^{\alpha\beta} \;=\; \begin{pmatrix} 0 & -E_1 & -E_2 & -E_3 \\ E_1 & 0 & -B_3 & B_2 \\ E_2 & B_3 & 0 & -B_1 \\ E_3 & -B_2 & B_1 & 0 \end{pmatrix} \;=\; \frac{\partial A^\beta}{\partial x_\alpha} - \frac{\partial A^\alpha}{\partial x_\beta} \tag{93}$$

Note that and $F^{\alpha\beta} = \partial^\alpha A^\beta - \partial^\beta A^\alpha$ is of the definition of the electromagnetic fields using the potentials

$$\vec{E} = -\nabla\Phi - \frac{1}{c}\frac{\partial \vec{A}}{\partial t} \qquad \text{and} \qquad \vec{B} = \nabla \times \vec{A}$$

in terms of 4-vectors

$$\frac{\partial}{\partial x_\alpha} = \left(\frac{\partial}{\partial x_0},\, -\nabla\right) \qquad \text{and} \qquad A^\beta = \left(\Phi,\, \vec{A}\right)$$

The equation of motion in covariant (4-vector) form is therefore

$$\frac{dp^\alpha}{d\tau} = \frac{q}{c} F^{\alpha\beta} U_\beta \qquad \text{or} \qquad \frac{dU^\alpha}{d\tau} = \frac{q}{mc} F^{\alpha\beta} U_\beta \tag{94}$$

or in the vector form

$$\frac{d\mathbf{p}}{d\tau} = \frac{q}{c}\,\mathbf{FgU} \qquad \text{or} \qquad \frac{d\mathbf{U}}{d\tau} = \frac{q}{mc}\,\mathbf{FgU} \tag{95}$$

10(d) Lorentz Transformation of the Field

For a Lorentz transformation $\mathbf{A}$ with $\mathbf{A}^T\mathbf{gA} = \mathbf{g}$, the transformation of 4-vectors such as $\mathbf{p}$ and $\mathbf{U}$ is

$$\mathbf{p}' = \mathbf{Ap}\,, \qquad \frac{d\mathbf{p}'}{d\tau} = \mathbf{A}\,\frac{d\mathbf{p}}{d\tau} \qquad \text{and} \qquad \mathbf{U}' = \mathbf{AU}\,, \qquad \frac{d\mathbf{U}'}{d\tau} = \mathbf{A}\,\frac{d\mathbf{U}}{d\tau}$$

Since

$$\frac{d\mathbf{p}'}{d\tau} \;=\; \mathbf{A}\frac{d\mathbf{p}}{d\tau} \;=\; \frac{q}{c}\,\mathbf{AFgU} \;=\; \frac{q}{c}\,\mathbf{AF}(\mathbf{A}^T\mathbf{gA})\mathbf{U} \;=\; \frac{q}{c}\,\mathbf{F}'\mathbf{gU}'$$

where $\mathbf{F}'$ is the Lorentz transformation of the field strength tensor

$$\mathbf{F}' = \mathbf{AFA}^T \tag{96}$$

or

$$F'^{\alpha\beta} = A_{\alpha\gamma} F^{\gamma\delta} A^T_{\delta\beta} = A_{\alpha\gamma} F^{\gamma\delta} A_{\beta\delta} = \frac{\partial x'^\alpha}{\partial x^\gamma}\, F^{\gamma\delta}\, \frac{\partial x'^\beta}{\partial x^\delta}$$

where $(x'_0\,,\,x'_1\,,\,x'_2\,,\,x'_3)^T = \mathbf{A}\,(x_0\,,\,x_1\,,\,x_2\,,\,x_3)^T$ or $x'^\alpha = A_{\alpha\beta}\,x^\beta$ *i.e.* $\partial x'^\alpha/\partial x^\beta = A_{\alpha\beta}$.

Example 10.1 Lorentz transformation with $\vec{\beta} \parallel \vec{e}_x$.

In this case the transformation matrix is

$$\mathbf{A} = \begin{pmatrix} \mathbf{A}_2 & \mathbf{0}_2 \\ \mathbf{0}_2 & \mathbf{I}_2 \end{pmatrix} \qquad \text{where} \qquad \mathbf{A}_2 = \gamma \begin{pmatrix} 1 & -\beta \\ -\beta & 1 \end{pmatrix} = \mathbf{A}_2^T\,,$$

$\mathbf{0}_2$ and $\mathbf{I}_2$ are the 2×2 zero and identity matrix, respectively. The field strength tensor can also be written into a block matrix,

$$\mathbf{F} = \begin{pmatrix} \mathbf{F}_1 & -\mathbf{F}_2 \\ \mathbf{F}_2^T & \mathbf{F}_3 \end{pmatrix}$$

where

$$\mathbf{F}_1 = \begin{pmatrix} 0 & -E_1 \\ E_1 & 0 \end{pmatrix}\,, \qquad \mathbf{F}_2 = \begin{pmatrix} E_2 & E_3 \\ B_3 & -B_2 \end{pmatrix}\,, \qquad \mathbf{F}_3 = \begin{pmatrix} 0 & -B_1 \\ B_1 & 0 \end{pmatrix}$$

The transformation of $\mathbf{F}$ can be calculated as

$$\mathbf{F}' = \mathbf{A}\mathbf{F}\mathbf{A}^T = \begin{pmatrix} \mathbf{A}_2\mathbf{F}_1 & -\mathbf{A}_2\mathbf{F}_2 \\ \mathbf{F}_2^T & \mathbf{F}_3 \end{pmatrix} \mathbf{A}^T = \begin{pmatrix} \mathbf{A}_2\mathbf{F}_1\mathbf{A}_2^T & -\mathbf{A}_2\mathbf{F}_2 \\ \mathbf{F}_2^T\mathbf{A}_2^T & \mathbf{F}_3 \end{pmatrix} = \begin{pmatrix} \mathbf{F}'_1 & -\mathbf{F}'_2 \\ \mathbf{F}'^T_2 & \mathbf{F}'_3 \end{pmatrix}$$

where

$$\begin{cases} \mathbf{F}'_1 = \mathbf{A}_2\mathbf{F}_1\mathbf{A}_2^T = \mathbf{F}_1 \\ \mathbf{F}'_3 = \mathbf{F}_3 \end{cases} \qquad \text{and} \qquad \mathbf{F}'_2 = \mathbf{A}_2\mathbf{F}_2 = \gamma \begin{pmatrix} E_2 - \beta B_3 & E_3 + \beta B_2 \\ B_3 - \beta E_2 & -B_2 - \beta E_3 \end{pmatrix}$$

The transformation of the field is therefore

$$\begin{cases} E'_1 &=& E_1 \\ E'_2 &=& E_2 - \beta B_3 \\ E'_3 &=& E_3 + \beta B_2 \end{cases} \qquad \text{and} \qquad \begin{cases} B'_1 &=& B_1 \\ B'_2 &=& B_2 + \beta E_3 \\ B'_3 &=& B_3 - \beta E_2 \end{cases}$$

Example 10.2 Lorentz transformation with $\vec{\beta}$ in an arbitrary direction.

In this case the Lorentz transformation matrix in Eq. (29) can be written as

$$\mathbf{A} = \left(\begin{array}{c|ccc} \gamma & -\gamma\beta_x & -\gamma\beta_y & -\gamma\beta_z \\ \hline -\gamma\beta_x & 1 + \dfrac{(\gamma-1)\beta_x^2}{\beta^2} & \dfrac{(\gamma-1)\beta_x\beta_y}{\beta^2} & \dfrac{(\gamma-1)\beta_x\beta_z}{\beta^2} \\ -\gamma\beta_y & \dfrac{(\gamma-1)\beta_x\beta_y}{\beta^2} & 1 + \dfrac{(\gamma-1)\beta_y^2}{\beta^2} & \dfrac{(\gamma-1)\beta_y\beta_z}{\beta^2} \\ -\gamma\beta_z & \dfrac{(\gamma-1)\beta_x\beta_z}{\beta^2} & \dfrac{(\gamma-1)\beta_y\beta_z}{\beta^2} & 1 + \dfrac{(\gamma-1)\beta_z^2}{\beta^2} \end{array}\right) = \begin{pmatrix} \gamma & -\gamma\vec{\beta} \\ -\gamma\vec{\beta}^T & \mathbf{A}_{22} \end{pmatrix} \tag{97}$$

where $\vec{\beta} = (\beta_1 \quad \beta_2 \quad \beta_3)$ and $\vec{\beta}^T = (\beta_1 \quad \beta_2 \quad \beta_3)^T$ are the row and column vector for $\vec{\beta}$, respectively, and

$$\mathbf{A}_{22} \;=\; \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} + \alpha \begin{pmatrix} \beta_1^2 & \beta_1\beta_2 & \beta_1\beta_3 \\ \beta_1\beta_2 & \beta_2^2 & \beta_2\beta_3 \\ \beta_1\beta_3 & \beta_2\beta_3 & \beta_3^2 \end{pmatrix} \;=\; \mathbf{I} + \alpha\mathbf{L} \tag{98}$$

with

$$\alpha = \frac{\gamma - 1}{\beta^2} = \frac{\gamma^2}{\gamma + 1} \tag{99}$$

and

$$\mathbf{L} \;=\; \begin{pmatrix} \beta_1\beta_1 & \beta_2\beta_1 & \beta_3\beta_1 \\ \beta_1\beta_2 & \beta_2\beta_2 & \beta_3\beta_2 \\ \beta_1\beta_3 & \beta_2\beta_3 & \beta_3\beta_3 \end{pmatrix} \;=\; \begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix} \begin{pmatrix} \beta_1 & \beta_2 & \beta_3 \end{pmatrix} \;=\; \vec{\beta}^T\vec{\beta} \tag{100}$$

Note that $\mathbf{A}^T = \mathbf{A}$ and $\mathbf{A}_{22}^T = \mathbf{A}_{22}$. The field strength tensor can also be written as

$$\mathbf{F} = \begin{pmatrix} 0 & -\vec{E} \\ \vec{E}^T & \mathbf{F}_{22} \end{pmatrix} \qquad \text{where} \qquad \mathbf{F}_{22} = \begin{pmatrix} 0 & -B_3 & B_2 \\ B_3 & 0 & -B_1 \\ -B_2 & B_1 & 0 \end{pmatrix} \tag{101}$$

and $\vec{E} = (E_1 \quad E_2 \quad E_3)$ and $\vec{E}^T = (E_1 \quad E_2 \quad E_3)^T$ are the row and column vector for electric field $\vec{E}$, respectively. The transformation of $\mathbf{F}$ is then

$$\begin{array}{ccccc} \mathbf{F}' & = & \mathbf{AFA}^T & = & \mathbf{AFA} \\ \| & & & & \| \end{array}$$

$$\begin{aligned} \begin{pmatrix} F'_{11} & -\vec{E}' \\ \vec{E}'^T & \mathbf{F}'_{22} \end{pmatrix} &= \begin{pmatrix} \gamma & -\gamma\vec{\beta} \\ -\gamma\vec{\beta}^T & \mathbf{A}_{22} \end{pmatrix} \begin{pmatrix} 0 & -\vec{E} \\ \vec{E}^T & \mathbf{F}_{22} \end{pmatrix} \begin{pmatrix} \gamma & -\gamma\vec{\beta} \\ -\gamma\vec{\beta}^T & \mathbf{A}_{22} \end{pmatrix} \\ &= \begin{pmatrix} \gamma\vec{\beta}\vec{E}^T & -\gamma\vec{E} - \gamma\vec{\beta}\,\mathbf{F}_{22} \\ \mathbf{A}_{22}\vec{E}^T & \gamma\vec{\beta}^T\vec{E} + \mathbf{A}_{22}\mathbf{F}_{22} \end{pmatrix} \begin{pmatrix} \gamma & -\gamma\vec{\beta} \\ -\gamma\vec{\beta}^T & \mathbf{A}_{22} \end{pmatrix} \end{aligned} \tag{102}$$

where $\vec{E}'$ and $\vec{E}'^T$ are the row and column vector for the electric field in K' frame, respectively, and $\mathbf{F}'_{22}$ is a 3×3 matrix. Note that $\mathbf{F}'$ is guaranteed to be an antisymmetric matrix, $\mathbf{F}'^T = -\mathbf{F}'$, since $\mathbf{A}^T = \mathbf{A}$ and $\mathbf{F}^T = -\mathbf{F}$. We now calculate each matrix element of $\mathbf{F}'$ from Eq. (102).

$$F'_{11} \;=\; -\gamma^2\vec{\beta}\,\vec{E}^T + \gamma^2(\vec{E} + \vec{\beta}\,\mathbf{F}_{22})\vec{\beta}^T \;=\; \gamma^2\,\vec{\beta}\,\mathbf{F}_{22}\,\vec{\beta}^T \;=\; \gamma^2\,\vec{\beta}\cdot(\vec{\beta}\times\vec{B}) \;=\; 0 \tag{103}$$

$$-\vec{E}' \;=\; \gamma^2\vec{\beta}\,\vec{E}^T\vec{\beta} - \gamma\,(\vec{E} + \vec{\beta}\,\mathbf{F}_{22})\mathbf{A}_{22} \;=\; \gamma^2(\vec{\beta}\cdot\vec{E})\,\vec{\beta} - \gamma\,(\vec{E} + \vec{\beta}\times\vec{B})\,\mathbf{A}_{22} \tag{104}$$

$$\mathbf{F}'_{22} \;=\; \gamma\,\vec{\beta}^T\vec{E}\,\mathbf{A}_{22} - \gamma\,\mathbf{A}_{22}\,\vec{E}^T\vec{\beta} + \mathbf{A}_{22}\,\mathbf{F}_{22}\,\mathbf{A}_{22} \tag{105}$$

where

$$\vec{\beta}\,\vec{E}^T = \vec{E}\,\vec{\beta}^T = \vec{\beta}\cdot\vec{E} \qquad \text{and} \qquad \vec{\beta}\,\mathbf{F}_{22} = \mathbf{F}_{22}^T\,\vec{\beta}^T = \vec{\beta}\times\vec{B}$$

Since $\mathbf{A}_{22} = \mathbf{I} + \alpha\mathbf{L} = \mathbf{I} + \alpha\vec{\beta}^T\vec{\beta}$, the 2nd and 3rd term in Eq. (104) are

$$\vec{E}\,\mathbf{A}_{22} \;=\; \vec{E} + \alpha\,(E_1 \;\; E_2 \;\; E_3)\begin{pmatrix}\beta_1\\ \beta_2\\ \beta_3\end{pmatrix}\begin{pmatrix}\beta_1 & \beta_2 & \beta_3\end{pmatrix} \;=\; \vec{E} + \frac{\gamma^2}{\gamma+1}(\vec{\beta}\cdot\vec{E}\,)\,\vec{\beta}$$

where $\alpha = \gamma^2/(\gamma+1)$, and

$$\begin{aligned}(\vec{\beta}\times\vec{B}\,)\,\mathbf{A}_{22} \;&=\; (\vec{\beta}\times\vec{B}\,) + \alpha\left((\vec{\beta}\times\vec{B})_1 \quad (\vec{\beta}\times\vec{B})_2 \quad (\vec{\beta}\times\vec{B})_3\right)\begin{pmatrix}\beta_1\\ \beta_2\\ \beta_3\end{pmatrix}\begin{pmatrix}\beta_1 & \beta_2 & \beta_3\end{pmatrix}\\ &=\; (\vec{\beta}\times\vec{B}\,) + \alpha[(\vec{\beta}\times\vec{B}\,)\cdot\vec{\beta}\,]\,\vec{\beta} \;=\; (\vec{\beta}\times\vec{B}\,)\end{aligned}$$

The transformation of the electric field is thus obtained from Eq. (104) as

$$\begin{aligned}\vec{E'} \;&=\; -\left[\gamma^2(\vec{\beta}\cdot\vec{E})\,\vec{\beta} - \gamma\,(\vec{E}+\vec{\beta}\times\vec{B})\,\mathbf{A}_{22}\right]\\ &=\; -\gamma^2(\vec{\beta}\cdot\vec{E})\,\vec{\beta} + \gamma\left[\vec{E} + \frac{\gamma^2}{\gamma+1}(\vec{\beta}\cdot\vec{E}\,)\,\vec{\beta} + (\vec{\beta}\times\vec{B}\,)\right]\\ &=\; \gamma\left(\vec{E}+\vec{\beta}\times\vec{B}\right) - \frac{\gamma^2}{\gamma+1}(\vec{\beta}\cdot\vec{E}\,)\,\vec{\beta}\end{aligned} \tag{106}$$

The transformation of the magnetic field can be obtained from Eq. (105),

$$\begin{aligned}\mathbf{F'}_{22} \;&=\; \gamma\left[\vec{\beta}^T\vec{E}\mathbf{A}_{22} - \mathbf{A}_{22}\vec{E}^T\vec{\beta}\right] + \mathbf{A}_{22}\mathbf{F}_{22}\mathbf{A}_{22}\\ &=\; \gamma\left[\left((\mathbf{I}+\alpha\mathbf{L})\,\vec{E}^T\vec{\beta}\right)^T - (\mathbf{I}+\alpha\mathbf{L})\,\vec{E}^T\vec{\beta}\right] + (\mathbf{I}+\alpha\mathbf{L})\,\mathbf{F}_{22}\,(\mathbf{I}+\alpha\mathbf{L})\\ &=\; \gamma\left[(\vec{E}^T\vec{\beta}\,)^T - \vec{E}^T\vec{\beta}\right] + \gamma\alpha\left[(\mathbf{L}\vec{E}^T\vec{\beta}\,)^T - \mathbf{L}\vec{E}^T\vec{\beta}\right]\\ &\qquad + \mathbf{F}_{22} + \alpha\,[\mathbf{F}_{22}\,\mathbf{L} + \mathbf{L}\,\mathbf{F}_{22}] + \alpha^2\mathbf{L}\,\mathbf{F}_{22}\,\mathbf{L}\end{aligned} \tag{107}$$

where $\mathbf{A}_{22} = \mathbf{I} + \alpha\mathbf{L} = \mathbf{I} + \alpha\vec{\beta}^T\vec{\beta}$. Since

$$\vec{E}^T\vec{\beta} \;=\; \begin{pmatrix}E_1\\ E_2\\ E_3\end{pmatrix}\begin{pmatrix}\beta_1 & \beta_2 & \beta_3\end{pmatrix} \;=\; \begin{pmatrix} & \vdots & \\ \cdots & E_i\beta_j & \cdots\\ & \vdots & \end{pmatrix},$$

the 1st bracket in Eq. (107) is

$$\begin{aligned}
\left[(\vec{E}^T\vec{\beta}\,)^T - \vec{E}^T\vec{\beta}\right] &= \begin{pmatrix} 0 & E_2\beta_1 - E_1\beta_2 & E_3\beta_1 - E_1\beta_3 \\ E_1\beta_2 - E_2\beta_1 & 0 & E_3\beta_2 - E_2\beta_3 \\ E_1\beta_3 - E_3\beta_1 & E_2\beta_3 - E_3\beta_2 & 0 \end{pmatrix} \\
&= \begin{pmatrix} 0 & (\vec{\beta}\times\vec{E})_3 & -(\vec{\beta}\times\vec{E})_2 \\ -(\vec{\beta}\times\vec{E})_3 & 0 & (\vec{\beta}\times\vec{E})_1 \\ (\vec{\beta}\times\vec{E})_2 & -(\vec{\beta}\times\vec{E})_1 & 0 \end{pmatrix}
\end{aligned} \tag{108}$$

Since

$$\vec{\beta}\vec{E}^T = \begin{pmatrix} \beta_1 & \beta_2 & \beta_3 \end{pmatrix} \begin{pmatrix} E_1 \\ E_2 \\ E_3 \end{pmatrix} = (\vec{\beta}\cdot\vec{E})$$

and

$$\begin{aligned}
\mathbf{L}\vec{E}^T\vec{\beta} &= \vec{\beta}^T\vec{\beta}\vec{E}^T\vec{\beta} = \vec{\beta}^T(\vec{\beta}\cdot\vec{E})\vec{\beta} = (\vec{\beta}\cdot\vec{E})\vec{\beta}^T\vec{\beta} = (\vec{\beta}\cdot\vec{E})\mathbf{L} \\
(\mathbf{L}\vec{E}^T\vec{\beta}\,)^T &= (\vec{\beta}\cdot\vec{E})\mathbf{L}^T = (\vec{\beta}\cdot\vec{E})\mathbf{L}
\end{aligned}$$

the 2nd bracket in Eq. (107) is

$$\left[(\mathbf{L}\vec{E}^T\vec{\beta}\,)^T - \mathbf{L}\vec{E}^T\vec{\beta}\right] = 0 \tag{109}$$

Since

$$\begin{aligned}
\mathbf{F}_{22}\mathbf{L} &= \mathbf{F}_{22}\,\vec{\beta}^T\vec{\beta} = (\vec{B}\times\vec{\beta})^T\vec{\beta} = \begin{pmatrix} & \vdots & \\ \cdots & (\vec{B}\times\vec{\beta})_i\,\beta_j & \cdots \\ & \vdots & \end{pmatrix} \\
\mathbf{L}\,\mathbf{F}_{22} &= -\,(\mathbf{F}_{22}\mathbf{L})^T
\end{aligned}$$

where $\mathbf{F}_{22}^T = -\mathbf{F}_{22}$, $\mathbf{L}^T = \mathbf{L}$, and

$$\mathbf{F}_{22}\,\vec{\beta}^T = \begin{pmatrix} 0 & -B_3 & B_2 \\ B_3 & 0 & -B_1 \\ -B_2 & B_1 & 0 \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix} = (\vec{B}\times\vec{\beta})^T\,,$$

The 3rd bracket in Eq. (107) is

$$\begin{aligned}
[\mathbf{F}_{22}\,\mathbf{L} + \mathbf{L}\,\mathbf{F}_{22}] &= \begin{pmatrix} & \vdots & \\ \cdots & (\vec{B}\times\vec{\beta})_i\,\beta_j - (\vec{B}\times\vec{\beta})_j\,\beta_i & \cdots \\ & \vdots & \end{pmatrix} \\
&= \begin{pmatrix} 0 & \left((\vec{B}\times\vec{\beta})\times\vec{\beta}\right)_3 & -\left((\vec{B}\times\vec{\beta})\times\vec{\beta}\right)_2 \\ -\left((\vec{B}\times\vec{\beta})\times\vec{\beta}\right)_3 & 0 & \left((\vec{B}\times\vec{\beta})\times\vec{\beta}\right)_1 \\ \left((\vec{B}\times\vec{\beta})\times\vec{\beta}\right)_2 & -\left((\vec{B}\times\vec{\beta})\times\vec{\beta}\right)_1 & 0 \end{pmatrix}
\end{aligned} \tag{110}$$

The last term in Eq. (107) is

$$\mathbf{L}\,\mathbf{F}_{22}\,\mathbf{L} \;=\; \vec{\beta}^T\vec{\beta}\,\mathbf{F}_{22}\,\vec{\beta}^T\vec{\beta} \;=\; \vec{\beta}^T\vec{\beta}\left(\vec{B}\times\vec{\beta}\right)^T\vec{\beta} \;=\; \vec{\beta}^T\left[\vec{\beta}\cdot(\vec{\beta}\times\vec{B}\,)\right]\vec{\beta} \;=\; 0 \qquad (111)$$

Combining Eqs. (108), (109), (110), and (111) yields the transformation of the magnetic field,

$$\begin{pmatrix} 0 & -B_3' & B_2' \\ B_3' & 0 & -B_1' \\ -B_2' & B_1' & 0 \end{pmatrix} = \mathbf{F}_{22} + \gamma\left[(\vec{E}^T\vec{\beta}\,)^T - \vec{E}^T\vec{\beta}\right] + \gamma\alpha\left[(\mathbf{L}\vec{E}^T\vec{\beta}\,)^T - \mathbf{L}\vec{E}^T\vec{\beta}\right] + \alpha\left[\mathbf{F}_{22}\,\mathbf{L} + \mathbf{L}\,\mathbf{F}_{22}\right] + \alpha^2\mathbf{L}\,\mathbf{F}_{22}\,\mathbf{L}$$

$$= \begin{pmatrix} 0 & -B_3 & B_2 \\ B_3 & 0 & -B_1 \\ -B_2 & B_1 & 0 \end{pmatrix} + \gamma\begin{pmatrix} 0 & (\vec{\beta}\times\vec{E})_3 & -(\vec{\beta}\times\vec{E})_2 \\ -(\vec{\beta}\times\vec{E})_3 & 0 & (\vec{\beta}\times\vec{E})_1 \\ (\vec{\beta}\times\vec{E})_2 & -(\vec{\beta}\times\vec{E})_1 & 0 \end{pmatrix}$$

$$+\frac{\gamma^2}{\gamma+1}\begin{pmatrix} 0 & \left((\vec{B}\times\vec{\beta})\times\vec{\beta}\right)_3 & -\left((\vec{B}\times\vec{\beta})\times\vec{\beta}\right)_2 \\ -\left((\vec{B}\times\vec{\beta})\times\vec{\beta}\right)_3 & 0 & \left((\vec{B}\times\vec{\beta})\times\vec{\beta}\right)_1 \\ \left((\vec{B}\times\vec{\beta})\times\vec{\beta}\right)_2 & -\left((\vec{B}\times\vec{\beta})\times\vec{\beta}\right)_1 & 0 \end{pmatrix}$$

Therefore,

$$\begin{aligned} \vec{B}' &= \vec{B} - \gamma(\vec{\beta}\times\vec{E}) - \frac{\gamma^2}{\gamma+1}\left[(\vec{B}\times\vec{\beta})\times\vec{\beta}\right] \\ &= \vec{B} - \gamma(\vec{\beta}\times\vec{E}) - \frac{\gamma^2}{\gamma+1}\left[(\vec{\beta}\cdot\vec{B})\vec{\beta} - \beta^2\vec{B}\right] \;=\; \gamma\left[\vec{B} - (\vec{\beta}\times\vec{E})\right] - \frac{\gamma^2}{\gamma+1}(\vec{\beta}\cdot\vec{B})\vec{\beta} \end{aligned}$$

In summary, the transformation of the field for $\vec{\beta}$ in an arbitrary direction is

$$\left\{\begin{aligned} \vec{E}\,' &= \gamma(\vec{E} + \vec{\beta}\times\vec{B}) - \frac{\gamma^2}{\gamma+1}\,(\vec{\beta}\cdot\vec{E})\,\vec{\beta} \\ \vec{B}\,' &= \gamma(\vec{B} - \vec{\beta}\times\vec{E}) - \frac{\gamma^2}{\gamma+1}\,(\vec{\beta}\cdot\vec{B})\,\vec{\beta} \end{aligned}\right. \qquad (112)$$

Electric and Magnetic Field from a Moving Charge

Consider a charge moving with a velocity $\vec{v}$. Let K' be the proper frame that moves with the charge and K be the Lab frame. In K' frame, the charge is stationary and the magnetic and electric fields produced by the charge are

$$\vec{B}\,' = 0 \qquad \text{and} \qquad \vec{E}\,' = \frac{kq}{r'^2}\left(\frac{\vec{r}\,'}{r'}\right) \qquad (113)$$

where $k = 1$ in the Gaussian unit. Note that $\vec{E}\,'$ is spherically symmetric in the proper frame. In the Lab frame,

$$\left\{ \begin{array}{lcl} \vec{E} & = & \gamma\vec{E}\,' - \dfrac{\gamma^2}{\gamma+1}\,(\vec{\beta}\cdot\vec{E}\,')\,\vec{\beta} \\ \vec{B} & = & \gamma(\vec{\beta}\times\vec{E}\,') \end{array} \right. \quad \Longrightarrow \quad \left\{ \begin{array}{lcl} E_{\parallel} & = & \gamma E'_{\parallel} - \dfrac{\gamma^2}{\gamma+1}\,(\vec{\beta}\cdot\vec{E}\,')\beta \\ \vec{E}_{\perp} & = & \gamma\vec{E}\,'_{\perp} \\ \vec{B} & = & \gamma(\vec{\beta}\times\vec{E}\,') \end{array} \right. \tag{114}$$

Note this is the transformation of Eq. (112) with $\vec{\beta} \longrightarrow -\vec{\beta}$, the inverse transformation from K' to K. In the lab frame, therefore, $\vec{E}$ is no longer spherically symmetric with the field being compressed in the direction of the motion. In the case of $\beta \sim 1$ and $\gamma >> 1$,

$$\left\{ \begin{array}{lclcl} \vec{E} & = & \gamma\left[\vec{E}\,' - (\vec{\beta}\cdot\vec{E}\,')\vec{\beta}\right] & = & \gamma\vec{E}\,'_{\perp} \\ \vec{B} & = & \gamma(\vec{\beta}\times\vec{E}\,') & \sim & \gamma\vec{E}\,'_{\perp} \end{array} \right.$$

For $\vec{\beta} = \pm\beta\,\vec{e}_x\,,\ \vec{E} = \gamma(0,\,E'_y,\,E'_z)$ and $\vec{B} = \pm\gamma(0,\,-E'_z,\,E'_y)\,.$

Invariance of $\vec{E}\cdot\vec{B}$ and $E^2 - B^2$

The determinant of the field strength tensor **F** is invariant under the Lorentz transformations because for any Lorentz transformation **A**,

$$\det(\mathbf{F}') \;=\; \det(\mathbf{A}\mathbf{F}\mathbf{A}^T) \;=\; [\det(\mathbf{A})]^2\det(\mathbf{F}) \;=\; \det(\mathbf{F})$$

Since the determinant of **F** can be calculated as

$$\begin{aligned} \det(\mathbf{F}) &= E_1\begin{vmatrix} E_1 & -B_3 & B_2 \\ E_2 & 0 & -B_1 \\ E_3 & B_1 & 0 \end{vmatrix} - E_2\begin{vmatrix} E_1 & 0 & B_2 \\ E_2 & B_3 & -B_1 \\ E_3 & -B_2 & 0 \end{vmatrix} + E_3\begin{vmatrix} E_1 & 0 & -B_3 \\ E_2 & B_3 & 0 \\ E_3 & -B_2 & B_1 \end{vmatrix} \\ &= E_1(E_1B_1^2 + E_2B_1B_2 + E_3B_3B_1) - E_2(-E_1B_1B_2 - E_2B_2^2 + E_3B_2B_3) \\ &\quad +E_3(E_1B_1B_3 + E_2B_2B_3 + E_3B_3^2) \\ &= (\vec{E}\cdot\vec{B})^2 \end{aligned}$$

$(\vec{E}\cdot\vec{B})$ is therefore invariant under the Lorentz transformation. In HW 11.14, $(E^2 - B^2)$ is shown to be another quadratic function of the field strength that is invariant for the Lorentz transformation.

10(e) Maxwell equation in Covariant Form

In vaccum, two inhomogeneous equations in Maxwell equations can be written as covariant form using the field strength tensor. For

$$\begin{cases} \nabla \cdot \vec{E} = 4\pi\rho \\ \nabla \times \vec{B} - \dfrac{1}{c}\dfrac{\partial \vec{E}}{\partial t} = \dfrac{4\pi}{c}\vec{J} \end{cases}$$

with $J^\alpha = (J_0 \;\; J_1 \;\; J_2 \;\; J_3) = (c\rho \;\; \vec{J})$, the inhomogeneous Maxwell equations can be written as

$$\begin{pmatrix} \partial_0 & \partial_1 & \partial_2 & \partial_3 \end{pmatrix} \begin{pmatrix} 0 & -E_1 & -E_2 & -E_3 \\ E_1 & 0 & -B_3 & B_2 \\ E_2 & B_3 & 0 & -B_1 \\ E_3 & -B_2 & B_1 & 0 \end{pmatrix} = \frac{4\pi}{c}\begin{pmatrix} J_0 & J_1 & J_2 & J_3 \end{pmatrix}$$

i.e.

$$(\partial_0 \;\; \partial_1 \;\; \partial_2 \;\; \partial_3)\,\mathbf{F} \;=\; \frac{4\pi}{c}(J_0 \;\; J_1 \;\; J_2 \;\; J_3) \qquad \text{or} \qquad \partial_\alpha F^{\alpha\beta} = \frac{4\pi}{c}J^\beta \tag{115}$$

For the covariant form of two homogeneous equations of Maxwell equations, we define a dual field strength tensor as

$$\mathcal{F}^{\alpha\beta} = \frac{1}{2}\epsilon^{\alpha\beta\gamma\delta}F_{\gamma\delta} = \begin{pmatrix} 0 & -B_1 & -B_2 & -B_3 \\ B_1 & 0 & E_3 & -E_2 \\ B_2 & -E_3 & 0 & E_1 \\ B_3 & E_2 & -E_1 & 0 \end{pmatrix} \tag{116}$$

where

$$\epsilon^{\alpha\beta\gamma\delta} = \begin{cases} 1 & \text{even permutation of } (\alpha,\beta,\gamma,\delta) \\ -1 & \text{odd permutation of } (\alpha,\beta,\gamma,\delta) \\ 0 & \text{any two indices are equal} \end{cases} \tag{117}$$

The change from field strength tensor to dual field-strength tensor is simply

$$\vec{E} \longrightarrow -\vec{B} \qquad \text{and} \qquad \vec{B} \longrightarrow -\vec{E} \qquad \Longrightarrow \qquad F^{\alpha\beta} \longrightarrow \mathcal{F}^{\alpha\beta}$$

The homogeneous equations can be written as covariant form as

$$\begin{cases} \nabla \cdot \vec{B} = 0 \\ \nabla \times \vec{E} + \dfrac{1}{c}\dfrac{\partial \vec{B}}{\partial t} = 0 \end{cases} \qquad \Longrightarrow \qquad \partial_\alpha \mathcal{F}^{\alpha\beta} = 0 \tag{118}$$

The covariant form of the homogeneous equations can also be expressed using field strength tensor as

$$\partial_\alpha F_{\beta\gamma} + \partial_\beta F_{\gamma\alpha} + \partial_\gamma F_{\alpha\beta} = 0$$

where $F_{\alpha\beta}$ is the covariant field strength tensor,

$$F_{\alpha\beta} = g_{\alpha\eta}\, F^{\eta\xi}\, g_{\xi\beta} = \begin{pmatrix} 0 & E_1 & E_2 & E_3 \\ -E_1 & 0 & -B_3 & B_2 \\ -E_2 & B_3 & 0 & -B_1 \\ -E_3 & -B_2 & B_1 & 0 \end{pmatrix} , \qquad g_{\alpha\eta} = \begin{pmatrix} 1 & 0 \\ 0 & -\mathbf{I}_{3\times 3} \end{pmatrix} \tag{119}$$

The difference between $F^{\alpha\beta}$ and $F_{\alpha\beta}$ is a reverse of the direction of $\vec{E}$, *i.e.*

$$\vec{E} \longrightarrow -\vec{E} \qquad \text{for} \qquad F^{\alpha\beta} \longrightarrow F_{\alpha\beta}$$

For the field in media, we define the field strength tensor as

$$G^{\alpha\beta} = \begin{pmatrix} 0 & -D_1 & -D_2 & -D_3 \\ D_1 & 0 & -H_3 & H_2 \\ D_2 & H_3 & 0 & -H_1 \\ D_3 & -H_2 & H_1 & 0 \end{pmatrix} \tag{120}$$

and a similar dual field strength tensor $\mathcal{F}^{\alpha\beta}$ with substituting $\vec{E} \to \vec{D}$ and $\vec{B} \to \vec{H}$. The Maxwell equation in media in covariant form is

$$\begin{cases} \partial_\alpha G^{\alpha\beta} = \dfrac{4\pi}{c} J^\beta \\ \partial_\alpha \mathcal{F}^{\alpha\beta} = 0 \end{cases} \tag{121}$$

11. Relativistic Equations for the Dynamics of Spin

11(a) Magnetic Dipole Moment Due to the Orbital Motion of Charges

For a localized current distribution $\vec{J}(\vec{r}\,)$, the magnetic dipole moment for the expansion of the vector potential is defined as (see Eq. (5.54) in Jackson's Book)

$$\vec{m} = \frac{1}{2c} \int [\vec{r} \times \vec{J}(\vec{r}\,)]\, d\vec{r} \tag{122}$$

where the magnetization or magnetic moment is defined as

$$\mathcal{M}(\vec{r}\,) = \frac{1}{2c} [\vec{r} \times \vec{J}(\vec{r}\,)] \tag{123}$$

For a current in a closed loop,

$$\vec{m} = \frac{I}{2c} \oint \vec{r} \times d\vec{l} = \frac{I}{c} \vec{S} \tag{124}$$

where the loop integral is through the current loop and $\vec{S}$ is total vector area of the loop. For a particle with charge q moving with velocity $\vec{v}(t)$, the current density due to the motion of the charge is

$$\vec{J}(\vec{r}\,) = q\vec{v}\, \delta(\vec{r} - \vec{r}(t)) \tag{125}$$

and the magnetic moment due to the motion of the charge is then

$$\vec{m} = \frac{q}{2c}\left[\vec{r}(t) \times \vec{v}(t)\right] = \frac{q}{2mc}\vec{L} \tag{126}$$

where $\vec{L} = m(\vec{r} \times \vec{v})$ is the angular momentum of the orbital motion of the particle. In general, the magnetic moment of a charged particle can be written as

$$\vec{m} = \left(\frac{ge}{2mc}\right)\vec{L} \tag{127}$$

where e is a negative number for the electron charge, $(ge/2mc)$ is the gyomagnetic ratio and $g = 1$ is the g factor. With an external magnetic field $\vec{B}$, the total force and torque of the magnetic force on a current distribution $\vec{J}(\vec{r})$ is (Eqs. (5.12) and (5.13) in Jackson's book)

$$\vec{F} = \frac{1}{c}\int \vec{J}(\vec{r}) \times \vec{B}(\vec{r})\, d^3\vec{r} \tag{128}$$

$$\vec{\tau} = \frac{1}{c}\int \vec{r} \times (\vec{J} \times \vec{B})\, d^3\vec{r} \tag{129}$$

For a localized current in a closed loop where the space variation of $\vec{B}$ is negligible, the torque can be written as (Eq. (5.71) in Jackson's book)

$$\vec{\tau} = \vec{m} \times \vec{B} \tag{130}$$

and the non-relativistic equation of motion for the orbital angular momentum is then

$$\frac{d\vec{m}}{dt} = \left(\frac{ge}{2mc}\right)\vec{m} \times \vec{B} \tag{131}$$

11(b) Magnetic Moment for the Electron Spin

Let $\vec{s}$ be the the vector for the electron spin. In order to explain the anomalous Zeeman effect, the Uhlenbeck-Goudschmidt hypothesis assumed that the relationship between the angular momentum of orbital motion and its magnetic moment can be used to define the magnetic moment of the electron spin,

$$\vec{\mu} = \frac{gq}{2mc}\,\vec{s} \tag{132}$$

where q is negative for electron and the g factor was found to be $g = 2.002$ for electron and $g = 5.59$ for proton. Similar to the orbital angular momentum of a charged particle in magnetic field, the equation of motion for the electron spin in a magnetic field should look like,

$$\frac{d\vec{s}}{dt} = \vec{\mu} \times \vec{B} = \left(\frac{gq}{2mc}\right)\vec{s} \times \vec{B}$$

This equation is, however, only correct when the electron is stationary orbitally, *i.e.* in the rest (proper) frame of electron. Therefore, it should be written as

$$\frac{d\vec{s}\,'}{d\tau} = \left(\frac{gq}{2mc}\right)\vec{s}\,' \times \vec{B}\,' \tag{133}$$

where $\vec{s}\,'$ and $\vec{B}\,'$ are the spin and the magnetic field in the proper frame, respectively, and the electric field is negligible in the proper frame, otherwise the particle cannot be stationary. In the proper frame, the particle is at rest instantaneously if the orbital motion has an acceleration. The spin $\vec{s}\,'$ in the proper frame is the intrinsic spin of a particle.

11(c) BMT Equation of Motion For the Spin

To obtain the equation of motion of the electron spin in the lab frame, we need to find a correct form of 4-vector for the spin and then transfer the equation of motion in the proper frame in Eq. (133) to the lab frame. Let S^α be the 4-vector of the spin and

$$\begin{cases} S'^\alpha = (\,0\ \ \vec{s}\,'\,)^T \ \ \text{is the spin in the rest frame where } S'^0 \text{ is assumed to be zero} \\ S^\alpha = (\,S^0\ \ \vec{s}\,)^T \ \ \text{is the spin in the Lab frame in which the electron is moving with } \vec{\beta} \\ \mathbf{A} \ \ \text{is the Lorentz transformation from the lab to the rest frame moving with } \vec{\beta} \end{cases}$$

where S'^0 is chosen to be zero in the rest frame because there is no any logical reason and experimental evidence for S'^0 to be nonzero (Pauli-Lubanski). The transformation of the spin 4-vector from the lab to the rest frame is simply

$$(\ 0\quad \vec{s}\,'\ )^T = \mathbf{A}\,(\ S^0 \quad \vec{s}\ )^T \qquad \Longrightarrow \qquad \begin{cases} 0 \ = \ \gamma(S^0 - \vec{\beta}\cdot\vec{s}\,) \\ \vec{s}\,' \ = \ \vec{s} + \dfrac{(\gamma-1)}{\beta^2}(\vec{\beta}\cdot\vec{s}\,)\vec{\beta} - \gamma\vec{\beta}S^0 \end{cases} \tag{134}$$

where the 0-component of this transformation yields a constraint for the spin in the lab frame,

$$S'^0 = \gamma(S^0 - \vec{\beta}\cdot\vec{s}\,) = 0 \qquad \Longrightarrow \qquad \mathbf{U}^T\mathbf{g}\mathbf{S} = 0 \qquad \text{or} \qquad U_\alpha S^\alpha = 0 \tag{135}$$

with $\mathbf{U} = (\gamma c\ \ \gamma\vec{v}\,)^T$ being the 4-velocity of the particle. The spin 4-vector in the Lab frame is thus perpendicular to the 4-velocity of the particle's orbital motion. Moreover, Eq. (135) provides a definition of the 0-component of the spin 4-vector in the lab frame,

$$S^0 = \vec{\beta}\cdot\vec{s} \tag{136}$$

To derive the constraint on the equation of motion for S^α due to the relationship between the components of S^α, we differentiate Eq. (135) with respect to τ

$$\mathbf{U}^T\mathbf{g}\,\frac{d\mathbf{S}}{d\tau} = -\frac{d\mathbf{U}^T}{d\tau}\,\mathbf{g}\mathbf{S} \qquad \text{or} \qquad U_\alpha\frac{dS^\alpha}{d\tau} = -\frac{dU_\alpha}{d\tau}S^\alpha \tag{137}$$

where $d\mathbf{U}^T/d\tau$ can be replaced with the equation of the orbital motion in an electric and magnetic field given in Eq. (95), *i.e.*

$$\frac{d\mathbf{U}}{d\tau} = \left(\frac{q}{mc}\right)\mathbf{F}\mathbf{g}\mathbf{U} \qquad \Longrightarrow \qquad \frac{d\mathbf{U}^T}{d\tau} = \left(\frac{q}{mc}\right)\mathbf{U}^T\mathbf{g}\mathbf{F}^T$$

where q is negative for the electron charge. Substituting this orbital equation into Eq. (137) yields

$$\mathbf{U}^T\mathbf{g}\,\frac{d\mathbf{S}}{d\tau} \;=\; -\left(\frac{q}{mc}\right)\mathbf{U}^T\mathbf{g}\mathbf{F}^T\mathbf{g}\mathbf{S} \;=\; \left(\frac{q}{mc}\right)\mathbf{U}^T\mathbf{g}\mathbf{F}\mathbf{g}\mathbf{S} \tag{138}$$

where $\mathbf{F}^T = -\mathbf{F}$ because the field strength tensor in is an antisymmetric matrix. This is a constraint on the equation of motion of the spin 4-vector $\mathbf{S}$ in the lab frame.

To construct a covariant equation of motion of the spin 4-vector in the lab frame, we first try a 4-vector equation that is the generalization of the spin equation in the rest frame. In the rest frame, the spin equation in Eq. (133) can be written into the 4-vector form as

$$\frac{d\vec{s}\,'}{d\tau} = \left(\frac{gq}{2mc}\right)\vec{s}\,'\times\vec{B}\,' \qquad\Longrightarrow\qquad \frac{d\mathbf{S}'}{d\tau} = \left(\frac{gq}{2mc}\right)\mathbf{F}'\mathbf{g}\mathbf{S}'$$

where in the rest frame $S'^0 = 0$ and $\vec{E}' = 0$ or electric field is negligible, otherwise the electron cannot be stationary. We thus first try

$$\frac{d\mathbf{S}}{d\tau} = \left(\frac{gq}{2mc}\right)\mathbf{F}\mathbf{g}\mathbf{S} \tag{139}$$

as the spin equation in the lab frame. This tryout equation is however inconsistent with the constraint in Eq. (138) because multiplying $\mathbf{U}^T\mathbf{g}$ to the both sides of Eq. (139) yields

$$\mathbf{U}^T\mathbf{g}\,\frac{d\mathbf{S}}{d\tau} = \left(\frac{gq}{2mc}\right)\mathbf{U}^T\mathbf{g}\mathbf{F}\mathbf{g}\mathbf{S} \tag{140}$$

which is different from Eq. (138) because $g/2 \neq 1$ for the electron spin. The difference between the tryout equation of the spin and the constraint in Eq. (138) of the spin motion in the lab frame is

$$\text{Discrepancy of Eq. (139) with Spin Constraint in Eq. (138)} = \left(\frac{gq}{2mc} - \frac{q}{mc}\right)\mathbf{U}^T\mathbf{g}\mathbf{F}\mathbf{g}\mathbf{S}$$

To resolve this inconsistency, we add an additional term to Eq. (139) that goes to zero at the rest frame and patches the difference between Eq. (138) and Eq. (140). This additional term should be

$$\text{Additional Term of Eq. (139)} = \left[\frac{(2-g)q}{2mc}\,\mathbf{U}^T\mathbf{g}\mathbf{F}\mathbf{g}\mathbf{S}\right]\left(\frac{1}{c^2}\mathbf{U}\right) \tag{141}$$

Because the 4-velocity of the particle in the lab frame is

$$\mathbf{U} = (\,\gamma c \quad \gamma c\vec{\beta}\,)^T \qquad \text{with} \qquad \mathbf{U}^T\mathbf{g}\mathbf{U} = c^2$$

and in the rest frame $\mathbf{U}'$ is

$$\left\{\begin{array}{rcccl} U'^0 &=& \gamma(U^0 - \vec{\beta}\cdot\vec{U}) &=& c \\ \vec{U}\,' &=& \vec{U} + \dfrac{(\gamma-1)}{\beta^2}(\vec{\beta}\cdot\vec{U})\vec{\beta} - \gamma\vec{\beta}U^0 &=& 0 \end{array}\right. \qquad \text{with} \qquad \mathbf{U}'^T\mathbf{g}\mathbf{F}' = 0$$

where $F'^{0\beta} = 0$ in the rest frame, this additional term satisfies the requirements as

$$
\begin{aligned}
(\text{Additional-Term}) &= \left[\frac{(2-g)q}{2mc}\left(\mathbf{U}'^T\mathbf{g}\mathbf{F}'\right)\mathbf{g}\,\mathbf{S}'\right]\left(\frac{1}{c^2}\mathbf{U}'\right) = 0\,, \quad \text{in the rest frame} \\
\mathbf{U}^T\mathbf{g}\,(\text{Additional-Term}) &= \left[\frac{(2-g)q}{2mc}\mathbf{U}^T\mathbf{g}\mathbf{F}\mathbf{g}\mathbf{S}\right]\left(\frac{1}{c^2}\mathbf{U}^T\mathbf{g}\mathbf{U}\right) \qquad \text{in the lab frame} \\
&= \left[\frac{(2-g)q}{2mc}\mathbf{U}^T\mathbf{g}\mathbf{F}\mathbf{g}\mathbf{S}\right]
\end{aligned}
$$

With the additional term in Eq. (141), the modified Eq. (139) that is the covariant equation of motion for a spin, called BMT equation, is

$$
\frac{d\mathbf{S}}{d\tau} = \left(\frac{gq}{2mc}\right)\mathbf{F}\mathbf{g}\mathbf{S} + \frac{(2-g)q}{2mc^3}\left(\mathbf{U}^T\mathbf{g}\mathbf{F}\mathbf{g}\mathbf{S}\right)\mathbf{U} \tag{142}
$$

or in the index form

$$
\frac{dS^\alpha}{d\tau} = \left(\frac{gq}{2mc}\right)F^{\alpha\beta}S_\beta + \frac{(2-g)q}{2mc^3}\left(U_\gamma F^{\gamma\beta}S_\beta\right)U^\alpha \tag{143}
$$

BMT equation satisfies the spin constraint in Eq. (138) and goes back to Eq. (133) in the rest frame. Note that the original BMT equation is covariant for the Lorentz transformation. In the lab frame,

$$
\mathbf{U} = (\,\gamma c \quad \gamma c\vec{\beta}\,)^T \qquad \text{and} \qquad \mathbf{S} = (\,S^0 \quad \vec{s}\,)^T \quad \text{with} \quad S^0 = \vec{\beta}\cdot\vec{s}
$$

For a charged particle in an electric and magnetic field, the two terms in BMT equation in Eq. (142) can be calculated as

$$
\begin{aligned}
\mathbf{F}\mathbf{g}\mathbf{S} &= \left(\,\vec{s}\cdot\vec{E} \quad (\vec{\beta}\cdot\vec{s})\vec{E} + \vec{s}\times\vec{B}\,\right)^T \\
\mathbf{U}^T\mathbf{g}\mathbf{F}\mathbf{g}\mathbf{S} &= \gamma c\left[\vec{s}\cdot(\vec{E}+\vec{\beta}\times\vec{B}\,) - (\vec{\beta}\cdot\vec{s})(\vec{\beta}\cdot\vec{E}\,)\right]
\end{aligned}
$$

Therefore, BMT equation can also be written in 3-vector form in the lab frame as

$$
\frac{d\vec{s}}{dt} = \frac{gq}{2\gamma mc}\left[(\vec{\beta}\cdot\vec{s})\vec{E} + \vec{s}\times\vec{B}\right] + \frac{\gamma(2-g)q}{2mc}\left[\vec{s}\cdot(\vec{E}+\vec{\beta}\times\vec{B}\,) - (\vec{\beta}\cdot\vec{s})(\vec{\beta}\cdot\vec{E}\,)\right]\vec{\beta} \tag{144}
$$

where $dt = \gamma d\tau$. Note that BMT equation cannot be solved unless the orbital motion of the particle, $\vec{\beta} = \vec{\beta}(t)$, is known. For a charged particle in an electric and magnetic field, the orbital motion needs to be solved from the 4-vector equation of the orbital motion with the Lorentz force,

$$
\frac{d\mathbf{U}}{d\tau} = \left(\frac{q}{mc}\right)\mathbf{F}\mathbf{g}\mathbf{U} \qquad \Longrightarrow \qquad
\left\{
\begin{aligned}
\frac{d\gamma}{dt} &= \frac{q}{mc}\left(\vec{\beta}\cdot\vec{E}\right) \\
\frac{d(\gamma\vec{\beta})}{dt} &= \frac{q}{mc}\left(\vec{E}+\vec{\beta}\times\vec{B}\right)
\end{aligned}
\right.
$$

which yield a 3-vector equation of the orbital motion

$$\frac{d\vec{\beta}}{dt} = \frac{q}{\gamma mc}\left[\vec{E} + \vec{\beta} \times \vec{B} - (\vec{\beta} \cdot \vec{E})\vec{\beta}\right] \tag{145}$$

The dynamics of the spin is therefore coupled with the orbital motion of a charged particle in an electric and magnetic field. It should be noted that the BMT equation is only valid when

$$\left| d\vec{\beta}/dt \right| << \beta \tag{146}$$

so that the adiabatic approximation is valid, in which the rest frame can be considered as an inertial frame instantaneously during a orbital motion with a slow acceleration, otherwise the rest frame is no longer an inertial frame. Moreover, BMT equation is the classical limit of Dirac equation that is the quantum equation for a particle with spin.

12. Kinetic Effect of Spin Dynamics

Thomas precession is a kinetic effect of BMT equation —– an effect of the kinetic orbital motion on the intrinsic spin that is the spin in the rest frame of a particle due to the term from the constraint in Eq. (137) and is not due to the interaction between the spin and external field that is an external torque. Moreover, the Thomas precession is the rotation of the intrinsic spin observed in the lab frame. To separate the kinetic effect of the orbital motion and the interaction of between the spin and external field, we first re-write MBT equation as

$$\begin{array}{cccccccc}
\dfrac{d\mathbf{S}}{d\tau} & = & \left(\dfrac{gq}{2mc}\right)\mathbf{FgS} & + & \dfrac{(2-g)q}{2mc^3}\left(\mathbf{U}^T\mathbf{gFgS}\right)\mathbf{U} & & & \\
 & = & \left(\dfrac{gq}{2mc}\right)\mathbf{FgS} & - & \left(\dfrac{gq}{2mc^3}\right)\left(\mathbf{U}^T\mathbf{gFgS}\right)\mathbf{U} & + & \left(\dfrac{q}{mc^3}\right)\left(\mathbf{U}^T\mathbf{gFgS}\right)\mathbf{U} \\
 & & \uparrow & & \uparrow & & \uparrow \\
\text{from} & & \dfrac{d\mathbf{S}'}{d\tau} & & \mathbf{U}^T\mathbf{g}\,\dfrac{d\mathbf{S}}{d\tau} & & \dfrac{d\mathbf{U}^T}{d\tau}\mathbf{gS}
\end{array}$$

where the 2nd and 3rd term are due to the discrepancy of Eq. (139) that is the spin dynamics from the rest frame with Eq. (138) that is the spin constraint from the equation of orbital motion. The 3rd term is thus from the equation of orbital motion to meet the constraint from the orbital motion and is therefore the kinetic effect of the orbital motion on the spin dynamics. Substituting the equation of orbital motion

$$\frac{d\mathbf{U}^T}{d\tau} = \left(\frac{q}{mc}\right)\mathbf{U}^T\mathbf{gF}^T = -\left(\frac{q}{mc}\right)\mathbf{U}^T\mathbf{gF} \tag{147}$$

into the "kinetic term", where $\mathbf{F}^T = -\mathbf{F}$, BMT equation can be also written as

$$\begin{array}{rcccl} \dfrac{d\mathbf{S}}{d\tau} & = & \dfrac{gq}{2mc}\left[\mathbf{FgS} - \dfrac{1}{c^2}\left(\mathbf{U}^T\mathbf{gFgS}\right)\mathbf{U}\right] & - & \dfrac{1}{c^2}\left(\dfrac{d\mathbf{U}^T}{d\tau}\mathbf{gS}\right)\mathbf{U} \\ & & \parallel & & \parallel \\ & = & \left(\dfrac{d\mathbf{S}}{d\tau}\right)_\perp & + & \left(\dfrac{d\mathbf{S}}{d\tau}\right)_\parallel \end{array} \tag{148}$$

where $(d\mathbf{S}/d\tau)_\perp$ is perpendicular to the velocity $\mathbf{U}$ while $(d\mathbf{S}/d\tau)_\parallel$ is parallel to the velocity as

$$\begin{array}{rcccl} \mathbf{U}^T\mathbf{g}\left(\dfrac{d\mathbf{S}}{d\tau}\right)_\perp & \sim & \mathbf{U}^T\mathbf{g}\left(\mathbf{FgS}\right) - \dfrac{1}{c^2}\left(\mathbf{U}^T\mathbf{gFgS}\right)\left(\mathbf{U}^T\mathbf{gU}\right) & = & 0 \\ \mathbf{U}^T\mathbf{g}\left(\dfrac{d\mathbf{S}}{d\tau}\right)_\parallel & = & \dfrac{1}{c^2}\left(\dfrac{d\mathbf{U}^T}{d\tau}\mathbf{gS}\right)\mathbf{U}^T\mathbf{gU} & = & \left(\dfrac{d\mathbf{U}^T}{d\tau}\mathbf{gS}\right) \end{array}$$

where $\mathbf{U}^T\mathbf{gU} = c^2$. Moreover, $(d\mathbf{S}/d\tau)_\perp$ is proportional to g that related the spin to its associated magnetic moment in Eq. (132) while $(d\mathbf{S}/d\tau)_\parallel$ is independent of g. Therefore, $(d\mathbf{S}/d\tau)_\perp$ is the interaction between the magnetic moment and external field, the torque on the spin motion that is perpendicular to the velocity of a particle, while $(d\mathbf{S}/d\tau)_\parallel$ is the kinetic effect from the orbital motion because it is to fulfill the constraint on the spin motion from the kinetic orbital motion.

13. Spin Precession and Spin-Orbit Interaction — Fine Structure Splitting of Atomic Energy Levels

For an electron orbiting around a nucleus in an atom in the lab frame where the nucleus is fixed, the electron experiences only an electric field from the nucleus with

$$\vec{E} = E(r)\hat{r} \qquad \text{and} \qquad \vec{B} = 0 \tag{149}$$

where $E(r) \sim 1/r^2$ is spherically symmetric and $\hat{r} = \vec{r}/r$. In the lab frame, the space components of BMT equation in Eq. (144) with only $\vec{E}$ field can be written as

$$\frac{d\vec{s}}{dt} = \frac{q}{2\gamma mc}\left\{g(\vec{\beta}\cdot\vec{s})\vec{E} + \gamma^2(2-g)\left[(\vec{s}\cdot\vec{E}\,) - (\vec{\beta}\cdot\vec{s})(\vec{\beta}\cdot\vec{E}\,)\right]\vec{\beta}\right\} \tag{150}$$

Note that $\vec{s}$ is the spin of a particle in the lab frame which is not the nature (intrinsic) spin that is the spin in the rest frame of a particle. In quantum mechanics for the energy spectrum of an atom, we need to know the time evolution and Hamiltonian of the intrinsic spin of a particle observed in the lab frame. We need therefore change the lab frame spin $\vec{s}$ in Eq. (150) to the intrinsic spin $\vec{s}\,'$ while keeping the equation of the spin motion of Eq. (150) in the lab frame, *i.e.* we are going to study how the **intrinsic spin that is the spin in the rest frame of a particle evolves in time as viewed from the laboratory frame when the particle is moving.** To derive the equation of the intrinsic spin $\vec{s}\,'(t)$ in the lab frame, we change the spin vector $\vec{s}$ in Eq. (150) to the intrinsic spin $\vec{s}\,'$ using the Lorentz transformation of the spin between the lab (K) to rest (K') frame,

$$\left\{\begin{array}{lllll} \vec{s}\,' & = & \vec{s} + \dfrac{(\gamma-1)}{\beta^2}(\vec{\beta}\cdot\vec{s})\vec{\beta} - \gamma\vec{\beta}S^0 & = & \vec{s} - \dfrac{\gamma}{\gamma+1}(\vec{\beta}\cdot\vec{s})\vec{\beta} \\ \vec{s} & = & \vec{s}\,' + \dfrac{(\gamma-1)}{\beta^2}(\vec{\beta}\cdot\vec{s}\,')\vec{\beta} + \gamma\vec{\beta}S'^0 & = & \vec{s}\,' + \dfrac{\gamma^2}{\gamma+1}(\vec{\beta}\cdot\vec{s}\,')\vec{\beta} \end{array}\right. \tag{151}$$

where $S^0 = (\vec{\beta}\cdot\vec{s})$ in the lab frame, $S'^0 = 0$ in the rest frame, and $\vec{\beta}$ is the orbital velocity of the particle in the lab frame with $\beta^2 = (\gamma^2-1)/\gamma^2$. The time derivative of the intrinsic spin $\vec{s}\,'$ in terms of lab time t can be calculated through the Lorentz transformation as

$$\begin{aligned} \frac{d\vec{s}\,'}{dt} &= \frac{d}{dt}\left[\vec{s} - \frac{\gamma}{\gamma+1}(\vec{\beta}\cdot\vec{s})\vec{\beta}\right] \qquad (152) \\ &= \frac{d\vec{s}}{dt} - (\vec{\beta}\cdot\vec{s})\vec{\beta}\,\frac{d}{dt}\left(\frac{\gamma}{\gamma+1}\right) - \frac{\gamma}{\gamma+1}\left[\left(\vec{s}\cdot\frac{d\vec{\beta}}{dt}\right)\vec{\beta} + (\vec{\beta}\cdot\vec{s})\frac{d\vec{\beta}}{dt} + \left(\vec{\beta}\cdot\frac{d\vec{s}}{dt}\right)\vec{\beta}\right] \end{aligned}$$

where

$$\frac{d}{dt}\left(\frac{\gamma}{\gamma+1}\right) = \frac{\gamma^3}{(\gamma+1)^2}\left(\vec{\beta}\cdot\frac{d\vec{\beta}}{dt}\right) \tag{153}$$

and each term of the substitution of $d\vec{s}/dt$ and $\vec{s}$ using BMT equation in Eq. (150) and the Lorentz transformation of the spin vector in Eq. (151) can be calculated as

$$\left\{\begin{array}{lll} (\vec{\beta}\cdot\vec{s}) & = & \left(1 + \dfrac{\gamma^2\beta^2}{\gamma+1}\right)(\vec{\beta}\cdot\vec{s}\,') \;=\; \gamma(\vec{\beta}\cdot\vec{s}\,') \\ (\vec{s}\cdot\vec{E}) & = & (\vec{s}\,'\cdot\vec{E}) + \dfrac{\gamma^2}{\gamma+1}(\vec{\beta}\cdot\vec{s}\,')(\vec{\beta}\cdot\vec{E}) \\ \left(\vec{s}\cdot\dfrac{d\vec{\beta}}{dt}\right) & = & \left(\vec{s}\,'\cdot\dfrac{d\vec{\beta}}{dt}\right) + \dfrac{\gamma^2}{\gamma+1}(\vec{\beta}\cdot\vec{s}\,')\left(\vec{\beta}\cdot\dfrac{d\vec{\beta}}{dt}\right) \\ \left(\vec{\beta}\cdot\dfrac{d\vec{s}}{dt}\right) & = & \dfrac{\gamma q}{2mc}\left\{\left(g-2\beta^2\right)(\vec{\beta}\cdot\vec{s})(\vec{\beta}\cdot\vec{E}) - \beta^2(g-2)(\vec{s}\cdot\vec{E})\right\} \\ & = & \dfrac{q}{2mc}\left\{\left[\gamma(g-2)+2\right](\vec{\beta}\cdot\vec{s}\,')(\vec{\beta}\cdot\vec{E}) - \beta^2\gamma(g-2)(\vec{s}\,'\cdot\vec{E})\right\} \end{array}\right. \tag{154}$$

Substituting Eqs. (150), (153), and (154) into Eq. (152) yields the equation of motion for the intrinsic spin in the lab time as

$$\begin{aligned}
\frac{d\vec{s}\,'}{dt} &= \frac{q}{2\gamma mc}\left\{\gamma g(\vec{\beta}\cdot\vec{s}\,')\vec{E}+\gamma^2(2-g)\left[(\vec{s}\,'\cdot\vec{E}\,)-\frac{\gamma}{\gamma+1}(\vec{\beta}\cdot\vec{s}\,')(\vec{\beta}\cdot\vec{E}\,)\right]\vec{\beta}\right\} \\
&\quad -\frac{\gamma^4}{(\gamma+1)^2}\left(\vec{\beta}\cdot\frac{d\vec{\beta}}{dt}\right)(\vec{\beta}\cdot\vec{s}\,')\vec{\beta} \qquad (155) \\
&\quad -\frac{\gamma}{\gamma+1}\left\{\left[\left(\vec{s}\,'\cdot\frac{d\vec{\beta}}{dt}\right)+\frac{\gamma^2}{\gamma+1}(\vec{\beta}\cdot\vec{s}\,')\left(\vec{\beta}\cdot\frac{d\vec{\beta}}{dt}\right)\right]\vec{\beta}+\gamma(\vec{\beta}\cdot\vec{s}\,')\frac{d\vec{\beta}}{dt}\right\} \\
&\quad -\frac{q}{2mc}\left(\frac{\gamma}{\gamma+1}\right)\left\{\left[\gamma(g-2)+2\right](\vec{\beta}\cdot\vec{s}\,')(\vec{\beta}\cdot\vec{E}\,)-\beta^2\gamma(g-2)(\vec{s}\,'\cdot\vec{E}\,)\right\}\vec{\beta} \\
&= \frac{q}{2mc}\left\{g(\vec{\beta}\cdot\vec{s}\,')\vec{E}-\left[\frac{2\gamma}{\gamma+1}(\vec{\beta}\cdot\vec{s}\,')(\vec{\beta}\cdot\vec{E}\,)+(g-2)(\vec{s}\,'\cdot\vec{E}\,)\right]\vec{\beta}\right\} \\
&\quad -\frac{\gamma}{\gamma+1}\left\{\left[\gamma^2(\vec{\beta}\cdot\vec{s}\,')\left(\vec{\beta}\cdot\frac{d\vec{\beta}}{dt}\right)+\left(\vec{s}\,'\cdot\frac{d\vec{\beta}}{dt}\right)\right]\vec{\beta}+\gamma(\vec{\beta}\cdot\vec{s}\,')\frac{d\vec{\beta}}{dt}\right\} \qquad (156)
\end{aligned}$$

With the equation orbital motion of a charged particle in an electric field, Eq. (145) with $\vec{B}=0$,

$$\frac{d\vec{\beta}}{dt}=\frac{q}{\gamma mc}\left[\vec{E}-(\vec{\beta}\cdot\vec{E}\,)\vec{\beta}\right] \qquad (157)$$

the equation of the intrinsic spin can be further reduced to

$$\begin{aligned}
\frac{d\vec{s}\,'}{dt} &= \frac{q}{mc}\left\{\frac{g}{2}(\vec{\beta}\cdot\vec{s}\,')\vec{E}-\left[\frac{\gamma}{\gamma+1}(\vec{\beta}\cdot\vec{s}\,')(\vec{\beta}\cdot\vec{E}\,)+\frac{(g-2)}{2}(\vec{s}\,'\cdot\vec{E}\,)\right]\vec{\beta}\right. \\
&\qquad -\frac{1}{\gamma+1}\left[\gamma^2(\vec{\beta}\cdot\vec{s}\,')(1-\beta^2)(\vec{\beta}\cdot\vec{E}\,)+(\vec{s}\,'\cdot\vec{E}\,)-(\vec{\beta}\cdot\vec{E}\,)(\vec{\beta}\cdot\vec{s}\,')\right]\vec{\beta} \\
&\qquad \left.-\frac{\gamma}{\gamma+1}(\vec{\beta}\cdot\vec{s}\,')\left[\vec{E}-(\vec{\beta}\cdot\vec{E}\,)\vec{\beta}\right]\right\} \\
&= \frac{q}{mc}\left(\frac{g}{2}-\frac{\gamma}{\gamma+1}\right)\left[(\vec{\beta}\cdot\vec{s}\,')\vec{E}-(\vec{s}\,'\cdot\vec{E}\,)\vec{\beta}\right] \\
&= \frac{q}{mc}\left(\frac{g}{2}-\frac{\gamma}{\gamma+1}\right)\left(\vec{\beta}\times\vec{E}\right)\times\vec{s}\,' \;=\; \gamma\left(\frac{g}{2}-\frac{\gamma}{\gamma+1}\right)\left(\vec{\beta}\times\frac{d\vec{\beta}}{dt}\right)\times\vec{s}\,' \qquad (158)
\end{aligned}$$

which can be written as

$$\frac{d\vec{s}\,'}{dt}=\vec{\omega}\times\vec{s}\,' \qquad (159)$$

This equation is in the form of the equation for a rotation of a vector with rotational frequency $\vec{\omega}$. In this case, the vector is a spin and the rotation of a spin is called precession. This equation of the spin precession is called Thomas's equation and $\vec{\omega}$ is the precession frequency

$$\vec{\omega}=\frac{q}{mc}\left(\frac{g}{2}-\frac{\gamma}{\gamma+1}\right)\left(\vec{\beta}\times\vec{E}\right)=\gamma\left(\frac{g}{2}-\frac{\gamma}{\gamma+1}\right)\left(\vec{\beta}\times\frac{d\vec{\beta}}{dt}\right) \qquad (160)$$

For an electron in atom, $\vec{E} = E(r)\hat{r}$,

$$\vec{\omega} = -\frac{qE(r)}{mcr}\left(\frac{g}{2} - \frac{\gamma}{\gamma+1}\right)\left(\vec{r}\times\vec{\beta}\right) = -\frac{qE(r)}{m^2c^2r}\left(\frac{g}{2} - \frac{\gamma}{\gamma+1}\right)\vec{L} \tag{161}$$

where

$$\vec{L} = mc(\vec{r}\times\vec{\beta}\,) \tag{162}$$

is the orbital angular momentum of the electron. Note that the electric field $E(r)$ in the precession frequency is treated as an average field along the electron trajectory or wave function, which is equivalent to the approximation of decoupling the spin equation from the equation of the orbital motion.

Spin-Precession Frequency and Orbital-Rotational Frequency of a Particle

For a particle moving in a uniform circular orbit, the equation of the orbital motion can be written as

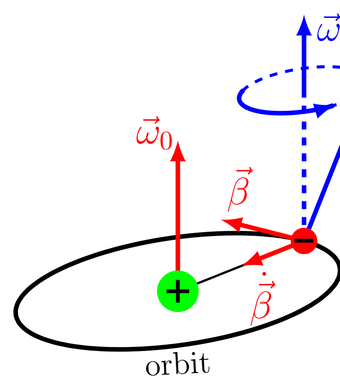
$$\frac{d\vec{\beta}}{dt} = \vec{\omega}_0 \times \vec{\beta} \tag{163}$$

where $\vec{\omega}_0$ is the angular velocity of the orbital motion. This equation of the uniform circular motion can be easily proved with the relationship between the velocity and angular velocity,

$$c\vec{\beta} = \vec{\omega}\times\vec{r} \qquad \Longrightarrow \qquad c\dot{\vec{\beta}} = \dot{\vec{\omega}}\times\vec{r} + \vec{\omega}\times\dot{\vec{r}}$$

For a uniform circular motion, $\dot{\vec{\omega}} = 0$, which yields Eq. (163). In a uniform circular orbit, the frequency of the spin precession of a particle is then

$$\begin{aligned}
\vec{\omega} &= \gamma\left(\frac{g}{2} - \frac{\gamma}{\gamma+1}\right)\left(\vec{\beta}\times\frac{d\vec{\beta}}{dt}\right) \\
&= \gamma\left(\frac{g}{2} - \frac{\gamma}{\gamma+1}\right)\left[\vec{\beta}\times\left(\vec{\omega}_0\times\vec{\beta}\right)\right] \\
&= \gamma\left(\frac{g}{2} - \frac{\gamma}{\gamma+1}\right)\beta^2\vec{\omega}_0 \\
&= \left[\frac{g(\gamma+1)}{2\gamma} - 1\right](\gamma-1)\,\vec{\omega}_0
\end{aligned} \tag{164}$$



Therefore,

$$\text{for } \beta << 1, \qquad \gamma \simeq 1 + \frac{1}{2}\beta^2 \qquad \Longrightarrow \qquad \omega \sim \frac{1}{2}\beta^2\,\omega_0$$

$$\text{for } \beta \sim 1, \qquad \gamma >> 1 \qquad \Longrightarrow \qquad \omega = \left(\frac{g}{2} - 1\right)\gamma\omega_0 \; >> \; \omega$$

Thomas Precession

Thomas precession is the rotation of the intrinsic spin observed in the lab frame due to the kinetic effect of the orbital motion. In spin equation (158), the term independent of g is from the kinetic effect. Therefore, the rotation of the intrinsic spin due to the acceleration of the orbital motion satisfies

$$\left(\frac{d\vec{s}\,'}{dt}\right)_{kinetic} \;=\; -\frac{\gamma^2}{\gamma+1}\left(\vec{\beta}\times\frac{d\vec{\beta}}{dt}\right)\times\vec{s}\,' \tag{165}$$

where the frequency of the Thomas precession is the term without g-factor of the total precession frequency in Eq. (164),

$$\vec{\omega}_T \;=\; -(\gamma-1)\vec{\omega}_0 \tag{166}$$

where, for the kinetic effect, the spin rotates in the opposite direction with respect to the orbital rotation of the particle because "$-$" sign in $\vec{\omega}_T$.

Hamiltonian for the Spin Precession of an Electron in Atom

From mechanics, the Hamiltonian for a uniform rotational motion with angular variable $\vec{\theta}$ and canonical conjugate momentum $\vec{L}$ is

$$\mathcal{H} = \vec{\omega}_0\cdot\vec{L} \qquad \text{for the orbital motion of } \frac{d\vec{\beta}}{dt} = \vec{\omega}_0\times\vec{\beta} \tag{167}$$

where the Hamilton's equation yields the uniform circular motion of

$$\frac{d\vec{\theta}}{dt} = \frac{\partial\mathcal{H}}{\partial\vec{L}} = \vec{\omega}_0 \qquad \text{and} \qquad \frac{d\vec{L}}{dt} = -\frac{\partial\mathcal{H}}{\partial\vec{\theta}} = 0 \tag{168}$$

For the spin of an electron in atom, the Hamiltonian for the spin precession is thus

$$\mathcal{H} \;=\; \vec{\omega}\cdot\vec{s}\,' \;\stackrel{\gamma\simeq 1}{=}\; \frac{(g-1)eE(r)}{2m^2c^2r}(\vec{L}\cdot\vec{s}\,') \;\sim\; \frac{1}{2}\beta^2\,\omega_0\,(\vec{L}\cdot\vec{s}\,') \tag{169}$$

where the precession frequency $\vec{\omega}$ is from Eq. (161) with $\gamma\simeq 1$ and $q=-e$ for electrons in atom. This is the Hamiltonian used for the spin-orbit interaction of electron in Hydrogen atom in quantum mechanics. The lowest energy state of the spin-orbit interaction is when $(\vec{L}\cdot\vec{s}\,') = -ls$, *i.e.* when $\vec{L}$ and $\vec{s}\,'$ are aligned in the opposite direction $(\uparrow\downarrow)$. The highest energy state is when $\vec{L}$ and $\vec{s}\,'$ are aligned in the same direction $(\uparrow\uparrow)$. Note that in Eq. (169), $\mathcal{H}_0 = \omega_0 L$ is the classical energy of the orbital motion of the electron. With $\beta << 1$ for the electron in Hydrogen atom, the energy of the spin precession is extremely small as compared with the energy of the orbital motion.

14. Spin Polarization of Particles Moving in Magnetic Field

If there is no electric field in the lab frame, $\vec{E}=0$, the BMT equation in the lab frame for the three space-components of a spin is

$$\frac{d\vec{s}}{dt} = \frac{gq}{2\gamma mc}\left\{\vec{s}\times\vec{B} - \gamma^2\left(1-\frac{2}{g}\right)\left[\vec{s}\cdot(\vec{\beta}\times\vec{B}\,)\right]\vec{\beta}\right\} \tag{170}$$

where the equation of the orbital motion is

$$\frac{d\vec{\beta}}{dt} = \frac{q}{\gamma mc}\left[\vec{E} + \vec{\beta}\times\vec{B} - (\vec{\beta}\cdot\vec{E})\vec{\beta}\right] \qquad \overset{\vec{E}=0}{\Longrightarrow} \qquad \frac{d\vec{\beta}}{dt} = \frac{q}{\gamma mc}\,(\vec{\beta}\times\vec{B}\,) \tag{171}$$

We now decompose $\vec{s}$ into the longitudinal and transverse components that are the spin polarized parallel and perpendicular to the direction of the orbital motion, respectively. Let

$$\vec{s} = (\vec{s}\cdot\hat{\beta})\hat{\beta} + \vec{s}\times\hat{\beta} = s_{\parallel}\,\hat{\beta} + \vec{s}_{\perp} \qquad \text{with} \quad \hat{\beta} = \vec{\beta}/\beta$$

where $s_{\parallel} = (\vec{s}\cdot\hat{\beta})$ is the longitudinal spin component and $\vec{s}_{\perp} = \vec{s}\times\hat{\beta}$ are the two transverse spin components on the transverse plane where $\hat{\beta}$ is the normal direction.

(a) Longitudinal Spin Component of a Particle Moving in Magnetic Field

The equation of motion for the longitudinal component can be obtained from

$$\frac{ds_{\parallel}}{dt} = \frac{d}{dt}(\vec{s}\cdot\hat{\beta}) = \frac{d\vec{s}}{dt}\cdot\hat{\beta} + \vec{s}\cdot\frac{d\hat{\beta}}{dt} = \frac{d\vec{s}}{dt}\cdot\hat{\beta} + \frac{q}{\gamma mc}\,\vec{s}\cdot(\hat{\beta}\times\vec{B}\,) \tag{172}$$

where, with $\vec{E} = 0$,

$$\frac{d\hat{\beta}}{dt} = \frac{d}{dt}\left(\frac{\vec{\beta}}{\beta}\right) = \frac{1}{\beta}\frac{d\vec{\beta}}{dt} - \frac{\vec{\beta}}{\beta^2}\frac{d\beta}{dt} = \frac{q}{\gamma mc}\,(\hat{\beta}\times\vec{B}\,) \tag{173}$$

with $d\beta/dt = 0$.

For the case of $\vec{E} \neq 0$ and $\vec{B} \neq 0$, the equation of the orbital motion is

$$\frac{d\gamma}{dt} = \frac{q}{mc}\left(\vec{\beta}\cdot\vec{E}\right) \qquad \text{and} \qquad \frac{d\vec{\beta}}{dt} = \frac{q}{\gamma mc}\left[\vec{E} + \vec{\beta}\times\vec{B} - (\vec{\beta}\cdot\vec{E})\vec{\beta}\right]$$

and

$$\frac{d\gamma}{dt} = \frac{\beta}{(1-\beta^2)^{3/2}}\frac{d\beta}{dt} \qquad \Longrightarrow \qquad \frac{d\beta}{dt} = \frac{1}{\beta\gamma^3}\frac{d\gamma}{dt} = \frac{q}{\beta\gamma^3 mc}\left(\vec{\beta}\cdot\vec{E}\right)$$

where $\gamma = 1/\sqrt{1-\beta^2}$. Equations (173) and (172) should be replaced with

$$\begin{aligned}
\frac{d\hat{\beta}}{dt} &= \frac{1}{\beta}\frac{d\vec{\beta}}{dt} - \frac{\vec{\beta}}{\beta^2}\frac{d\beta}{dt} = \frac{q}{\beta\gamma mc}\left[\vec{E} + \vec{\beta}\times\vec{B} - (\vec{\beta}\cdot\vec{E})\vec{\beta}\right] - \frac{q}{\beta^3\gamma^3 mc}\left(\vec{\beta}\cdot\vec{E}\right)\vec{\beta} \\
&= \frac{q}{\beta\gamma mc}\left[\vec{E} + \vec{\beta}\times\vec{B} - \frac{1}{\beta^2}\left(\vec{\beta}\cdot\vec{E}\right)\vec{\beta}\right]
\end{aligned} \tag{174}$$

and

$$\begin{aligned}
\frac{ds_{\parallel}}{dt} &= \frac{d\vec{s}}{dt}\cdot\hat{\beta} + \vec{s}\cdot\frac{d\hat{\beta}}{dt} \\
&= \frac{1}{\beta}\left(\frac{d\vec{s}}{dt}\cdot\vec{\beta}\right) + \frac{q}{\beta\gamma mc}\left[\left(\vec{s}\cdot\vec{E}\right) + \vec{s}\cdot\left(\vec{\beta}\times\vec{B}\right) - \frac{1}{\beta^2}\left(\vec{\beta}\cdot\vec{E}\right)\left(\vec{s}\cdot\vec{\beta}\right)\right]
\end{aligned} \tag{175}$$

With $\vec{E} = 0$, substituting the BMT equation in Eq. (170) into Eq. (172) yields the equation of motion for $s_\parallel$ as

$$\begin{aligned} \frac{ds_\parallel}{dt} &= \frac{(g-2)\gamma q}{2mc}\left[\vec{s}\cdot(\vec{B}\times\hat{\beta}\,)\right] = \frac{(g-2)\gamma q}{2mc}\left[(s_\parallel\,\hat{\beta}+\vec{s}_\perp\,)\cdot(\vec{B}\times\hat{\beta}\,)\right] \\ &= \frac{(g-2)\gamma q}{2mc}\left[(\vec{B}\times\hat{\beta}\,)\cdot\vec{s}_\perp\right] \end{aligned} \tag{176}$$

where $g = 2.002$ for electron and $g = 5.59$ for proton. For an electron with certain percentage of longitudinal and transverse spin components, $\vec{s} = s_\parallel\,\hat{\beta}+\vec{s}_\perp$, the change of the longitudinal spin is slow since $(g-2) = 0.002$ and, therefore, the longitudinal spin can be preserved for a relatively long period of time. For protons, on the other hand, $(g-2) = 3.59$, the longitudinal spin is not a constant of motion if there is a small amount of transverse spin components $\vec{s}_\perp$.

(b) Transverse Spin Components of a Particle Moving in Magnetic Field

The transverse component of the spin can be written as $\vec{s}_\perp = \vec{s}\times\hat{\beta}$. With Eqs. (170) and (173), the equation of motion for the transverse spin components can be obtained as

$$\begin{aligned} \frac{d\vec{s}_\perp}{dt} &= \frac{d\vec{s}}{dt}\times\hat{\beta}+\vec{s}\times\frac{d\hat{\beta}}{dt} = \frac{gq}{2\gamma mc}(\vec{s}\times\vec{B}\,)\times\hat{\beta}+\frac{q}{\gamma mc}\vec{s}\times(\hat{\beta}\times\vec{B}\,) \\ &= \frac{q}{2\gamma mc}\left[(g-2)s_\parallel\vec{B}-g(\hat{\beta}\cdot\vec{B}\,)(s_\parallel\,\hat{\beta}+\vec{s}_\perp)+2(\vec{s}\cdot\vec{B}\,)\hat{\beta}\right] \end{aligned} \tag{177}$$

If a particle moves in a plane and the magnetic field is along the normal direction of the plane, $(\hat{\beta}\cdot\vec{B}\,) = 0$, $(\vec{s}\cdot\vec{B}\,) = (\vec{s}_\perp\cdot\vec{B}\,)$ and

$$\frac{d\vec{s}_\perp}{dt} = \frac{q}{2\gamma mc}\left[(g-2)s_\parallel\vec{B}+2(\vec{s}_\perp\cdot\vec{B}\,)\hat{\beta}\right] \tag{178}$$

Note that $d\vec{s}_\perp/dt$ has a longitudinal component because the longitudinal-transverse coordinate is a coordinate moving along the particle trajectory. The transverse component of the spin is not conserved even the spin is initially polarized transversely.

Extra HW 1. Derive Eq. (11.98) in textbook —– the transformation matrix $\mathbf{A}$ between two inertial frames with $\vec{v}$ in arbitrary direction.

Extra HW 2. Prove the Lorentz transformation satisfies $\mathbf{A}^T\mathbf{g}\mathbf{A} = \mathbf{g}$.

Extra HW 3. Use the matrix A for the Lorentz transformation to derive the velocity transformation between two inertial reference frames moving in an arbitrary direction with respect to each other.

Extra HW 4. Use the formula for the velocity transformation between two inertial reference frames (from Extra HW 2) to verify that $U_0^2-\vec{U}\cdot\vec{U}$ is invariant under the Lorentz transformation.

Extra HW 5. Is it possible to have $\vec{E}' \perp \vec{B}'$ in an inertial frame and $\vec{E}\cdot\vec{B} \neq 0$ in another inertial frame?

Extra HW 6. Calculate force between two ultra-relativistic electrons moving in (a) the same direction and (b) the opposite direction.

Extra HW 7. Consider a uniform charge distribution on a spherical shell of radius r_0 and mass m. The spherical shell rotates with an angular velocity $\vec{\omega}$.

(a) Calculate the magnetic dipole moment of the shell.

(b) Obtain the relationship between the magnetic moment and the angular momentum of the shell.

Extra HW 8. For a particle moving in a central-force field in two dimensional space where the force on the particle is along the radial direction,

$$\vec{F} = -\nabla U(r) \qquad \text{with} \quad r = \sqrt{x^2 + y^2}$$

(a) Derive the condition for the circular orbit.

(b) Obtain the Hamiltonian in the action-angle variables when the particle is in the circular orbit and also obtain the canonical transformation from (x, p_x, y, p_y) to the action-angle variables.

Extra HW 9. (This problem is the similar as the HW 12.12 on the textbook.) Consider a muon beam circulates in a circular particle storage ring around a uniform magnetic field $\vec{B}$. An additional constant electric field $\vec{E}$ is also used around the ring for the focusing of the particle beam.

(a) Derive the equation of motion of the longitudinal component of the spin of the particles when both electric and magnetic field are included,

$$\frac{ds_\parallel}{dt} = \frac{q\gamma}{mc}\left[\left(\frac{g}{2} - 1\right)\vec{s}_\perp \cdot (\vec{B} \times \hat{\beta}\,) + \left(\frac{1}{\beta} - \frac{g\beta}{2}\right)\vec{s}_\perp \cdot \vec{E}\right]$$

(b) The storage ring can be operated at a "magic γ" at which the effect of the electric field on the spin polarization cancels out. Find the relationship between this magic γ and g of muon. This is a nuclear physics experiment to measure the value of g in very high precision.

Assignment 1. 11.1, 11.3, 11.4, 11.6

Assignment 2. Extra 1, Extra 2, Extra 3, Extra 4

Assignment 3. 11.23, 11.22

Assignment 4. 11.9, 11.10, 11.11

Assignment 5. 11.13, 11.14, Extra 5

Assignment 6. 11.15, Extra 6

Assignment 7. Extra 7, Extra 9